



Flexi Master Series Mock - 10 (2020)

Scorecard (procview.jsp?sid=aaa4FcNtTAH1R6G8XMNExTue Feb 16 12:08:27 IST 2021&qsetId=M5n8Po7Vhel=&qsetName=Flexi Master Series Mock - 10 (2020))

Accuracy (AccSelectGraph.jsp?sid=aaa4FcNtTAH1R6G8XMNExTue Feb 16 12:08:27 IST 2021&qsetId=M5n8Po7Vhel=&qsetName=Flexi Master Series Mock - 10 (2020))

Qs Analysis (QsAnalysis.jsp?sid=aaa4FcNtTAH1R6G8XMNExTue Feb 16 12:08:27 IST 2021&qsetId=M5n8Po7Vhel=&qsetName=Flexi Master Series Mock - 10 (2020))

Video Attempt (VideoAnalysis.jsp?sid=aaa4FcNtTAH1R6G8XMNExTue Feb 16 12:08:27 IST 2021&qsetId=M5n8Po7Vhel=&qsetName=Flexi Master Series Mock - 10 (2020))

Solutions (Solution.jsp?sid=aaa4FcNtTAH1R6G8XMNExTue Feb 16 12:08:27 IST 2021&qsetId=M5n8Po7Vhel=&qsetName=Flexi Master Series Mock - 10 (2020))

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VARC

LRDI

QA

Sec 1

Direction for questions (1-5): Read the given passage and answer the questions that follow.

[...] Despite what the haters might think, all areas of science confront questions that can't be answered within the process of science itself. Whenever scientists examine the best way to test a theory, or wonder how scientific models relate to reality, they're doing philosophy. But in its unique position as the study of the whole of existence, cosmology in particular is full of philosophical puzzles and positions.

In fact, there's a philosophical belief hiding at the very heart of cosmology. The *cosmological principle* states that, on large scales, the Universe is homogeneous (looks the same at all locations) and isotropic (looks the same in all directions). For example, the view from a ship in the middle of the ocean would be isotropic but, when land is in sight, the view is not the same in all directions. The ocean surface itself is homogeneous, perhaps, until you get near the shore.

The cosmological principle is fundamental to our understanding of how the Universe evolved, expanding from a uniform, hot plasma and cooling to form the intricate cosmic web we can now see through our telescopes. To assume homogeneity and isotropy everywhere, one must first average over insignificant, smaller differences, such as whole planets and even galaxies. The cosmological principle is thus a *statistical* principle: it is true only if you apply it to large-enough scales.

But even then, it might not be true. The Universe need not be homogeneous; Albert Einstein's theory of gravity works just fine if it isn't, and gravity causes structures to grow over time, exaggerating tiny initial differences. Whether these initial differences came from the 'quantum fluctuations' of virtual particles popping into and out of existence is unresolved.

So scientists are left in a state of hesitant acceptance. The cosmological principle is foundational to how we describe the evolution of the Universe, yet so far we've been unable to prove that it's necessarily true. Attempts to measure whether the Universe is homogeneous – or at what scale it becomes homogeneous – have met with mixed results. But cosmological *isotropy* has indeed been observed: the Cosmic Microwave Background radiation, emitted from everywhere in the Universe a few hundred thousand years after the Big Bang, is isotropic to one part in 100,000. Analogously, our ship in the ocean might see tiny differences, such as little choppy waves, but the view is largely isotropic.

Now, it's possible to get isotropy without homogeneity. To an observer in the centre of a spherical distribution of matter, things look the same in all directions, but such a distribution need not be homogeneous. However, many cosmologists are content to believe that homogeneity at some scale exists, whether or not it's been measured – because with the help of a non-empirical, philosophical principle, homogeneity logically follows from isotropy.

This is known as the Copernican principle, which states that there are no privileged observers – we are not in a special place in the Universe, and the centre is certainly a very special place. By this principle, the Universe must be isotropic everywhere, from all vantage points and not just ours – and in order for that to be true, the Universe must be homogeneous as well. If every ship sees a view that looks isotropic, there must be no land to make things look any different, so the ocean must be the same at each location. [...]

Q.1 [11594329]

Why is the cosmological principle a statistical principle?

1 ☐ Because it is not true if the sample size is not negligible.

2 ☐ Because it is applicable to large scale samples.

3 ☐ Because it is meaningless for gigantic data sizes.

4 ☐ Because it is not applicable to quantum physics.



Solution:

Correct Answer : 2

Your Answer : 2

Genre: Philosophy

Word Count# 557

 Answer key/Solution

Refer to the line: "The cosmological principle is thus a *statistical* principle: it is true only if you apply it to large-enough scales." Option 2 is the correct answer.

Option 1 – 'Not negligible' may refer to small sets of data too. It doesn't clearly answer the question.

Option 3 – It is factually wrong.

Option 4 – This is irrelevant.

Bookmark

FeedBack

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Q.2 [11594329]

In this passage, the author mainly tries to:

-
- 1 ☐ resolve the paradox between the Copernican principle and the cosmological principle.

2 ☐ highlight the contrast between philosophical theories and scientific studies.

3 ☐ prove the haters of Philosophical pursuit wrong.

4 ☐ explain the philosophical basis of cosmological study.



Solution:

Correct Answer : 4

Your Answer : 4

Genre: Philosophy

Word Count# 557

Answer key/Solution

The main idea of the author is that philosophy is relevant to the understanding of cosmology.

Option 1 – There is no paradox. It is also too narrow an option to be the main idea.

Option 2 – The author tries to show similarities or interdependence between the two disciplines. So, it is an incorrect option.

Option 3 – The author refers to haters in the first paragraph. There is not enough data to define these. This option is also too narrow.

Option 4 – It is the best choice.

Bookmark

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Q.3 [11594329]

Why is cosmology especially related to Philosophy?

1 ☐ Because it asks puzzles and posits questions.

2 ☐ Because Science is inherently related to Philosophy and Cosmology is the main branch of Science.

3 ☐ Because both deal with questions related to the entirety of existence.

4 ☐ Because Cosmology is opposed by the proponents of Philosophy.



Solution:

Correct Answer : 3

Your Answer : 3

Genre: Philosophy

Word Count# 557

 Answer key/Solution

Refer to the line: "But in its unique position as the study of the whole of existence, cosmology in particular is full of philosophical puzzles and positions." Option 3 is the clear answer. 'Entirety of existence' is 'whole of existence'.

Option 1 – It is a vague option. Philosophical puzzles don't necessarily mean all puzzles.

Option 2 – There is no data given in the passage related to this option.

Option 4 – This is factually wrong.

Bookmark

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Q.4 [11594329]

Which of the following can be inferred about the Copernican principle?

-
- 1 ☐ It proves that the Earth is the centre of the cosmos.

- 2 ☐ It helps prove the logical correlation between homogeneity and isotropy.
- 3 ☐ It proves that something that appears homogeneous may in fact not be isotropic.
- 4 ☐ It helps prove the theory that everything isotropic must originally be homogeneous.



Solution:

Correct Answer : 2

Your Answer : 4

Genre: Philosophy

Word Count# 557

 Answer key/Solution

The author talks about Copernican principle in the last paragraph. It is mentioned to show how homogeneity is logically supposed to result in isotropy. So, option 2 is the answer.

Option 1 – This is factually not supported by the passage. Secondly, a finding of Copernicus can't be an inference.

Option 3 – This goes against the discussion of the author.

Option 4 – The reverse is true. Homogeneity leads to isotropy, not the other way round. Secondly, this is not a proven conclusion.

Bookmark

FeedBack

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Q.5 [11594329]

Which of the following views would not be isotropic, as per the author?

- 1 ☐ A child staring at the first floor of his home from the ground below.

2 ☐ A man stranded in the middle of a desert with no path in view.

3 ☐ A girl sitting in the middle of a huge dark room that has no door or window.

4 ☐ A boy looking at the sky from inside a plane.



Solution:

Correct Answer : 1

Your Answer : 3

Genre: Philosophy

Word Count# 557

 **Answer key/Solution**

As per the passage, isotropy is defined as “looks the same in all directions”. Options 2, 3, and 4 will all look the same due to the range of the landscape. Option 1 may or may not be isotropic as all the rooms in the floor may not be identical. So, option 1 is the answer.

Bookmark

FeedBack

Direction for questions (6-10): Read the given passage and answer the questions that follow.

Anorexia is the third most chronic illness among adolescents. It leads more often to death than any other mental illness. Therapies that have the greatest success involve extensive family engagement, with parents undertaking the task of persuading the child to eat. It's exhausting and terrifying and demanding for parents, especially if recovery is slow.

Now a new study from Columbia University has identified for the first time what's going on in the brain when anorexics make a decision about what to eat. By using fMRIs, the researchers found that when they decide what food to eat, people with anorexia are engaging a part of the brain that is associated with the habitual control of actions, rather than the part of the brain which is associated with values.

This suggests that anorexics don't decide what to eat or not to eat because they need to lose or gain weight; the brain has just gotten into the habit of deciding not to eat. "The part of the brain they're using is different from the part of the brain healthy people seem to be using," says Daphna Shohamy, a neuroscientist and one of the authors of the study.

While the study was small—just 21 participants with anorexia and 21 without—and did not set out to look for treatment options, there were some implications for parents struggling with an anorexic child. "When we talk about the study with parents it makes an awful lot of sense to them," says Dr. Joanna Steinglass, who specializes in anorexia and worked on the study along with Karin Foerde of New York University and Timothy Walsh of Columbia. "They see habits as they get laid down. They see the behaviour become entrenched. They see behaviour becoming hard to change. What we've now done is begun to look at what basic mechanisms would be that might explain it."

One of the clear implications seems to be that family-based therapy is more likely to work than just talk therapy, since it appears to be the behaviour that needs to change, not any given set of beliefs, or how a patient is feeling that day. "Treatments are going to have to focus in on the behaviour in order to be successful," says Dr. Steinglass.

The study suggests that parents might want to try various strategies to get the kid's brain out of the habit of deciding to eat very little food. It's possible, says Dr. Steinglass that the cue that sets off the behaviour could be changed. "Before [the anorexia sufferers] walk into the meal, before they pick up the fork and knife, what could you do instead of what you always do that might shake it up a little?"

She tells the story of a parent at a conference whose child cut up their food into very small pieces before eating it. One way to short-circuit such behaviour, she suggests, might be to get the child to eat with only his or her left hand. "It's a very over simplified but useful illustration," Dr. Steinglass says. "If you try eating with your left hand, does that just slow the whole meal down enough that you can pay attention? It helps raise awareness of what you're doing."

Another way the study might help parents is in addressing this as a brain problem and not a willpower problem. "Parents seem to find helpful to understand that something has happened where behaviours have gotten stuck," says Dr. Steinglass. "It's not really about logical thinking at this point."

Q.6 [11594329]

Why does the author give the example of the strategy of asking an anorexic child to eat with only one hand?

- 1 ☐ In order to show that the child won't be able to cut the food into small pieces.
- 2 ☐ In order to explain how the child will be cured of anorexia.
- 3 ☐ In order to show how the parents can stop the child from developing bad habits.
- 4 ☐ In order to explain the benefit of parental involvement in curing anorexia.



Solution:

Correct Answer : 4

Your Answer : 2

Genre: Psychology

Word Count# 586

 Answer key/Solution

The author gives the example while talking about the possible application of the findings of the study. The author states that understanding the role of brain in curing anorexia will help the parents with a possible strategy. Then the example about not being able to cut food is given. So, option 4 is the clear answer.

Option 1 – This is not the purpose of the example. We need to look at the bigger picture. It is too narrow an explanation.

Option 2 – This is a method to cure anorexia. But we can't state with certainty that by doing this one thing the child will be cured. So, this is an extreme option. Option 4 is a better choice.

Option 3 – 'Bad habits' is a very broad term. So, this option is too generic.

Bookmark

FeedBack

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Q.7 [11594329]

Which of the following has been admitted by the author as a drawback of the study cited in the passage?

- 1 ☐ The slow recovery rate can frustrate parents who have anorexic kids.
- 2 ☐ The small sample size and lack of focus on the treatment of the illness.
- 3 ☐ The association between value and habit in terms of brain functions is vague.
- 4 ☐ The lack of any relevant findings for parents struggling with anorexic children.



Solution:

Correct Answer : 2

Your Answer : 1

Genre: Psychology

Word Count# 586

 Answer key/Solution

Refer to the part, “While the study was small—just 21 participants with anorexia and 21 without—and did not set out to look for treatment options...”. So, option 2 is the clear answer. The other options can be easily eliminated.

Option 1 – It’s not a drawback of the study. It is a problem with the entire treatment process of anorexia.

Options 3 and 4 – Irrelevant and incorrect respectively.

Bookmark

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Q.8 [11594329]

As per the passage, what can be inferred about the association between brain and anorexia?

- 1 ☐ The difference between an anorexic and non-anorexic is the difference between habits and values.
- 2 ☐ An anorexic brain can't develop new habits.
- 3 ☐ A non-anorexic person knows how to distinguish between good and bad habits.
- 4 ☐ The brain of an anorexic treats not eating food differently than that of a non-anorexic person.



Solution:

Correct Answer : 4

Your Answer : 1

Genre: Psychology

Word Count# 586

 Answer key/Solution

Refer to the part: "By using fMRIs, the researchers found that when they decide what food to eat, people with anorexia are engaging a part of the brain that is associated with the habitual control of actions, rather than the part of the brain which is associated with values." So, option 4 is a valid inference.

Option 1 – 'Habits' and 'values' are broad terms. This option is too generic a conclusion to be supported by the passage.

Option 2 – We can't surely say that. A habit can be good or bad. An anorexic person may be able to develop new habits.

Option 3 – The passage is about anorexic people. Any inference about non-anorexic people will be outside the scope of the passage, and, hence, illogical.

Bookmark

FeedBack

Direction for questions (6-10): Read the given passage and answer the questions that follow.

Anorexia is the third most chronic illness among adolescents. It leads more often to death than any other mental illness. Therapies that have the greatest success involve extensive family engagement, with parents undertaking the task of persuading the child to eat. It's exhausting and terrifying and demanding for parents, especially if recovery is slow.

Now a new study from Columbia University has identified for the first time what's going on in the brain when anorexics make a decision about what to eat. By using fMRIs, the researchers found that when they decide what food to eat, people with anorexia are engaging a part of the brain that is associated with the habitual control of actions, rather than the part of the brain which is associated with values.

This suggests that anorexics don't decide what to eat or not to eat because they need to lose or gain weight; the brain has just gotten into the habit of deciding not to eat. "The part of the brain they're using is different from the part of the brain healthy people seem to be using," says Daphna Shohamy, a neuroscientist and one of the authors of the study.

While the study was small—just 21 participants with anorexia and 21 without—and did not set out to look for treatment options, there were some implications for parents struggling with an anorexic child. "When we talk about the study with parents it makes an awful lot of sense to them," says Dr. Joanna Steinglass, who specializes in anorexia and worked on the study along with Karin Foerde of New York University and Timothy Walsh of Columbia. "They see habits as they get laid down. They see the behaviour become entrenched. They see behaviour becoming hard to change. What we've now done is begun to look at what basic mechanisms would be that might explain it."

One of the clear implications seems to be that family-based therapy is more likely to work than just talk therapy, since it appears to be the behaviour that needs to change, not any given set of beliefs, or how a patient is feeling that day. "Treatments are going to have to focus in on the behaviour in order to be successful," says Dr. Steinglass.

The study suggests that parents might want to try various strategies to get the kid's brain out of the habit of deciding to eat very little food. It's possible, says Dr. Steinglass that the cue that sets off the behaviour could be changed. "Before [the anorexia sufferers] walk into the meal, before they pick up the fork and knife, what could you do instead of what you always do that might shake it up a little?"

She tells the story of a parent at a conference whose child cut up their food into very small pieces before eating it. One way to short-circuit such behaviour, she suggests, might be to get the child to eat with only his or her left hand. "It's a very over simplified but useful illustration," Dr. Steinglass says. "If you try eating with your left hand, does that just slow the whole meal down enough that you can pay attention? It helps raise awareness of what you're doing."

Another way the study might help parents is in addressing this as a brain problem and not a willpower problem. "Parents seem to find helpful to understand that something has happened where behaviours have gotten stuck," says Dr. Steinglass. "It's not really about logical thinking at this point."

Q.9 [11594329]

Which of the following is not untrue as per the passage?

- 1 ☐ Timothy Walsh is a Colombian.
- 2 ☐ Karin Foerde has no relation with Columbia University.
- 3 ☐ Joanna Steinglass is acquainted with Daphna Shohamy.
- 4 ☐ Daphna Shohamy was one of the subjects of the study mentioned in the passage.



Solution:

Correct Answer : 3

Your Answer : 1

Genre: Psychology

Word Count# 586

 Answer key/Solution

We need to find a statement that is true (not untrue).

Option 1 – He teaches at Columbia University. We have no information regarding his nationality. Refer to the part “Dr. Joanna Steinglass, who specializes in anorexia and worked on the study along with Karin Foerde of New York University and Timothy Walsh of Columbia...”

Option 2 – Karin Foerde is mentioned as one of the authors of the study. So, she does have a professional association with Columbia as the study was conducted by that university.

Option 3 – This is true. Daphna Shohamy is one of the co-authors of the study. So, she must know her colleagues. This is the answer.

Option 4 – She is mentioned as a co-author of the study. So, this is untrue too.

Bookmark

FeedBack

Direction for questions (6-10): Read the given passage and answer the questions that follow.

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Q.10 [11594329]

With which of the following would the author least likely agree?

1 ☐ A mental illness can be fatal.

2 ☐ A strong willpower is essential to fight anorexia.

3 ☐ An anorexic can change his/her attitude towards eating with therapies involving family engagement.

4 ☐ A slow rate of recovery can demotivate caregivers.

×

Solution:

Correct Answer : 2

Your Answer : 3

Genre: Psychology

Word Count# 586

[🔍 Answer key/Solution](#)

Option 1 – The author states that anorexia is a mental illness (first paragraph) and it kills a lot of people. So, the author would agree with this assertion. Note the use of the modal 'can'. It provides that logical cushion needed to make an inference logical.

Option 2 – In the last paragraph, the author clearly states that anorexia has more to do with the functioning of the brain, rather than willpower. So, the author would surely not agree with this. Hence, it is the answer.

Option 3 – It matches the main idea of the passage. The author also mentions family therapy. So, the author would agree with this.

Option 4 – Refer to the first paragraph. The author talks about slow recovery process being exhaustive. Parents are caregivers. So, this is a valid inference too.

Bookmark

FeedBack

Direction for questions (11-14): Read the given passage and answer the questions that follow.

[...] Small-ranged species are disproportionately at risk of extinction. The “Red List” of the International Union for the Conservation of Nature (IUCN) aspires to assess the risks of extinction of all species. It has done so for more than 90,000 plant and animal species, out of a total of nearly 2 million species that taxonomists have described. Birds, mammals, and amphibians are the best known. Overwhelmingly, it’s the species with small geographical ranges that are at greatest risk. For birds, the risk of extinction drops dramatically as range size increases. About half the species with ranges smaller than 1,000 square kilometres are at risk.

This pattern is not surprising. Other things being equal, the destruction of habitats is more likely to terminate a species that occurs, for instance, on a few mountaintops in coastal Brazil, than one that occurs across the entire Amazon basin. The challenge for conservation science is what we can do to protect these threatened species.

We now know that species with small geographical ranges, which are so often threatened, are concentrated in certain places. And in these mostly tropical concentrations, human actions not only destroy habitats but also leave what’s behind in small, isolated fragments. These patches may be too small to sustain viable populations of species. Restoring corridors—habitat connections between fragments—affords a cost-effective solution. [...]

Many more species with small geographical ranges are known now than in Darwin and Wallace’s time. By 1850, taxonomists had described close to 5,000 terrestrial bird species in the world. [...] In 1850, 14 years after Darwin returned from his voyage, only 200 (4 percent) of the known species at that time had ranges smaller than 10,000 square kilometres. Today, that number is 1,290—close to 13 percent of the more than 10,000 species now known.

We’re still finding more such species. Taxonomists have described half of the known amphibian and mammalian species with small ranges in the past few decades. Many more are surely awaiting discovery. As with birds, large-ranged species were discovered earlier than small-ranged ones.

The century and a half of exploration since the travels of Darwin and Wallace has made it possible to produce three key maps that show the areas that have, respectively, the greatest numbers of species, the greatest numbers of species with small ranges, and the greatest numbers of species that are threatened with extinction. Geography is destiny. Understanding the relationships between these geographical patterns is the first vital step in determining where to act to save species.

The greatest numbers of species in a given place are in the tropical moist forests of the world. The patterns are broadly similar for birds, mammals, and amphibians—the taxa we know best. Less geographically resolved data for insects and plants suggest the patterns are also broadly similar. Wallace went to the Amazon because that’s where the most species are! He made his living collecting novelties—species other collectors had not discovered. He likely thought that more species would mean more novelties—but he was partly wrong. [...]

The environmentalist Norman Myers coined the term “*biodiversity hotspots*” for these places where concentrations of small-ranged species have collided with extensive habitat loss. His influential insight was that human actions are exceptionally damaging. For various reasons, we have disproportionately harmed those places where small-ranged species are concentrated. Maps provide high-resolution improvements on his ground-breaking ideas. [...]

Q.11 [11594329]

As per the author, what is the most significant challenge for conservation science?

- 1 ☐ To identify the exact range of species that are under threat.
- 2 ☐ To find measures to contain the habitats of all animals at risk.
- 3 ☐ To protect the habitats of threatened species with a small geographical range.
- 4 ☐ To restore the lost habitats of all extinct species.



Solution:

Correct Answer : 3

Your Answer : 3

Genre: Biology / Ecology

Word Count# 561

Answer key/Solution

Refer to the line: "The challenge for conservation science is what we can do to protect these threatened species." Option 3 is the clear answer.

Option 1 – It is not mentioned even as a challenge.

Option 2 – 'All animals' makes it less important. The passage focuses on the species with the greatest risk: species with small habitat range.

Option 4 – Too broad and 'lost habitats of extinct species' doesn't make sense.

Bookmark

FeedBack

Direction for questions (11-14): Read the given passage and answer the questions that follow.

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This pattern is not surprising. Other things being equal, the destruction of habitats is more likely to terminate a species that occurs, for instance, on a few mountaintops in coastal Brazil, than one that occurs across the entire Amazon basin. The challenge for conservation science is what we can do to protect these threatened species.

We now know that species with small geographical ranges, which are so often threatened, are concentrated in certain places. And in these mostly tropical concentrations, human actions not only destroy habitats but also leave what’s behind in small, isolated fragments. These patches may be too small to sustain viable populations of species. Restoring corridors—habitat connections between fragments—affords a cost-effective solution. [...]

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Q.12 [11594329]

As per the passage, to sustain viable populations of species, humans should:

- 1 ☐ create new corridors.
- 2 ☐ restore destroyed corridors.
- 3 ☐ protect large habitats.
- 4 ☐ spread species across vast lands.



Solution:

Correct Answer : 2

Your Answer : 2

Genre: Biology / Ecology

Word Count# 561

[🔍 Answer key/Solution](#)

Refer to the lines: “These patches may be too small to sustain viable populations of species. Restoring corridors—habitat connections between fragments—affords a cost-effective solution.” Option 2 is the clear answer. The other options can be easily eliminated.

Bookmark

FeedBack

Direction for questions (11-14): Read the given passage and answer the questions that follow.

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Q.13 [11594329]**Why does the author call Geography as destiny?**

- 1 ☐ Because Geographical patterns play the most significant role in identifying species and their habitats.
- 2 ☐ Because without Geography, there won't be any maps to be produced.
- 3 ☐ Because geographical maps help the greatest numbers of species with small ranges.
- 4 ☐ Because understanding geographical patterns will help conserve the threatened species.

**Solution:****Correct Answer : 4****Your Answer : 4****Genre: Biology / Ecology****Word Count# 561**[🔍 Answer key/Solution](#)

Refer to the lines: "Geography is destiny. Understanding the relationships between these geographical patterns is the first vital step in determining where to act to save species." Option 4 best captures the essence of this statement.

Option 1 – 'Most significant' makes this option extreme. Secondly, identifying geographical patterns is not a concern raised by the author.

Option 2 – This is a very narrow interpretation and it has no bearing with the main idea of the passage.

Option 3 – This is misleading. Maps don't help. Humans need to interpret the maps to find patterns and then take measures that will help. So, it is a distorted option.

[Bookmark](#)[FeedBack](#)

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Q.14 [11594329]

Which of the following best describes the author's intention behind talking about 'biodiversity hotspots'?

- 1 ☐ To explain a possible step in protecting threatened species.
- 2 ☐ To show how this term helped create high-resolution maps.
- 3 ☐ To highlight a possible solution for creating new cost effective habitats.
- 4 ☐ To acknowledge the roles of Darwin, Wallace, and Myers in conservation science.



Solution:

Correct Answer : 1

Your Answer : 3

Genre: Biology / Ecology

Word Count# 561

 Answer key/Solution

The author mentions this in the last paragraph. The main intention of the author is to support his/her main idea in the passage. Option 1 provides the most direct match for this.

Option 2 – Irrelevant

Option 3 – 'New habitats' is not the main concern; restoring lost habitats is.

Option 4 – Irrelevant

Bookmark

FeedBack

Direction for questions (15-17): Read the given passage and answer the questions that follow.

A New York businessman who lives in East Orange and is something of a pigeon fancier recently lost several of his finest birds through the depredations of vagrant cats. A few days ago the losses became so heavy that he armed himself with a gun and lay in ambush one afternoon when he returned from the city.

After a wait he saw a lean cat emerge from the cote with one of his finest pigeons in its mouth. He fired, and the cat fell dead. In the early transports of his joy at having destroyed the thief he forgot that there was yet a task for him to perform, but soon recollected that the body must be disposed of. First he thought of digging a hole in the back yard and interring the cat therein, but then he trembled when he thought what the neighbours might think he was burying. At last a bright idea struck him.

"I'll wrap the cat in papers and throw it off the ferryboat when I cross in the morning," he promised himself.

So, with the bundle neatly tied, he took the train on the following morning. He got off the train and boarded the boat, and there he was greeted by a group of friends from whom he could not escape. He reflected that he might have to make embarrassing explanations if he threw the bundle overboard while he was with them, and he deferred the act until the boat landed, thinking he could easily cast it away in an ash barrel on the way to the office.

He passed several ash barrels on his way, but somehow or other some one always seemed to be gazing in his direction when he approached one, and once or twice he saw a watchful policeman. He recollected how unpleasant discoveries had been made in ash barrels, and he didn't want to be arrested on suspicion. So he went all the way to the office and carefully locked the body in a closet, reflecting he could throw it overboard on his way home.

Going across the river that night he met some more sociable acquaintances, and the cat boarded the train with him as a result. He laid the package down beside him and tried to become absorbed in his paper, but that everlasting cat haunted him. When he reached his station he picked up a package and went home. Reaching there, he handed the bundle to the cook and, as indifferently as he could, told her to bury the cat in the back yard.

"Yes, Sir," said the woman.

There were a few minutes of relief for the East Orangeite, but soon the cook reappeared.

"I guess there's some mistake, Sir. This isn't a cat in the paper. It's a nice leg of mutton."

The man had evidently picked up the wrong bundle on leaving the train, and he only hopes the other fellow who reached home with the dead cat doesn't learn his identity.

Q.15 [11594329]

Why was the man unable to throw the dead cat into an ash barrel?

1 ☐ He was scared of policemen.

2 ☐ He didn't want to cause a scene.

3 ☐ He couldn't do it without being noticed.

4 ☐ He couldn't find a suitable barrel.



Solution:

Correct Answer : 3

Your Answer : 3

Genre: Abstract / Humour

Word Count# 499

Answer key/Solution

Refer to the line: "He passed several ash barrels on his way, but somehow or other some one always seemed to be gazing in his direction when he approached one, and once or twice he saw a watchful policeman." Option 3 is the correct answer.

Option 1 – We can't say that he was, in general, scared of policemen.

Option 2 – Causing a scene and not being seen as suspicious are two different things.

Option 4 – This is factually incorrect. He passed through a lot of barrels. He couldn't find a suitable opportunity.

Bookmark

FeedBack

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Q.16 [11594329]

What best describes the order of events in the passage?

-
- 1 ☐ A man discovers the source of his problem; he commits a blunder in solving the problem; he tries to avoid embarrassment; he ends up with yet another embarrassment.
-
- 2 ☐ A man solves a problem; he finds himself in a dilemma; he tries to find a resolution; he commits a blunder.
-

3 ○ A man identifies and solves a source of his problems; he tries to get rid of the source; he commits one blunder after another.

4 ○ A man kills a cat; he tries to avoid disposing the body of the cat; he ends up giving the dead cat to an unsuspecting bystander.

Solution:

Correct Answer : 2

Genre: Abstract / Humour

Word Count# 499

 Answer key/Solution

Option 2 is the only factually correct option.

Option 1 – He didn't commit any blunder in killing the cat. He doesn't end up with an embarrassing situation. He ends up with a blunder.

Option 3 – 'One blunder after another' is too extreme and doesn't match the storyline.

Option 4 – He doesn't give the dead cat to anyone. So, this is incorrect too.

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Direction for questions (15-17): Read the given passage and answer the questions that follow.

A New York businessman who lives in East Orange and is something of a pigeon fancier recently lost several of his finest birds through the depredations of vagrant cats. A few days ago the losses became so heavy that he armed himself with a gun and lay in ambush one afternoon when he returned from the city.

After a wait he saw a lean cat emerge from the cote with one of his finest pigeons in its mouth. He fired, and the cat fell dead. In the early transports of his joy at having destroyed the thief he forgot that there was yet a task for him to perform, but soon recollected that the body must be disposed of. First he thought of digging a hole in the back yard and interring the cat therein, but then he trembled when he thought what the neighbours might think he was burying. At last a bright idea struck him.

"I'll wrap the cat in papers and throw it off the ferryboat when I cross in the morning," he promised himself.

So, with the bundle neatly tied, he took the train on the following morning. He got off the train and boarded the boat, and there he was greeted by a group of friends from whom he could not escape. He reflected that he might have to make embarrassing explanations if he threw the bundle overboard while he was with them, and he deferred the act until the boat landed, thinking he could easily cast it away in an ash barrel on the way to the office.

He passed several ash barrels on his way, but somehow or other some one always seemed to be gazing in his direction when he approached one, and once or twice he saw a watchful policeman. He recollected how unpleasant discoveries had been made in ash barrels, and he didn't want to be arrested on suspicion. So he went all the way to the office and carefully locked the body in a closet, reflecting he could throw it overboard on his way home.

Going across the river that night he met some more sociable acquaintances, and the cat boarded the train with him as a result. He laid the package down beside him and tried to become absorbed in his paper, but that everlasting cat haunted him. When he reached his station he picked up a package and went home. Reaching there, he handed the bundle to the cook and, as indifferently as he could, told her to bury the cat in the back yard.

"Yes, Sir," said the woman.

There were a few minutes of relief for the East Orangeite, but soon the cook reappeared.

"I guess there's some mistake, Sir. This isn't a cat in the paper. It's a nice leg of mutton."

The man had evidently picked up the wrong bundle on leaving the train, and he only hopes the other fellow who reached home with the dead cat doesn't learn his identity.

Q.17 [11594329]

What is the tone of the author in the last sentence of the passage?

1 ☐ Deprecating

2 ☐ Condescending

3 ☐ Appealing4 ☐ Sarcastic

✕

Solution:**Correct Answer : 4****Your Answer : 3****Genre: Abstract / Humour****Word Count# 499**[🔍 Answer key/Solution](#)

The tone in the last line is humorous. So, option 4 is the right answer. Options 1 and 2 are too extreme and negative. Option 3 is irrelevant to this context.

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Q.18 [11594329]

Directions for question (18): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. This effect becomes important when the acceleration due to gravity falls below a certain threshold, as it does in the outer reaches of galaxies.
2. If $f(R)$ theories try to do away with dark energy, theories of modified Newtonian dynamics (MOND) do away with dark matter.
3. Effectively, gravity no longer weakens with distance, explaining why stars and gas in the outskirts of galaxies orbit just as fast as they do near the core.
4. In this approach, objects respond differently to gravity than Isaac Newton envisaged.

Solution:**Correct Answer : 2413****Sentences 1 and 4 can't open the paragraph as each has the pronoun 'this'.****Sentence 3 starts with 'effectively' which shows that it adds to another sentence.****Thematically, it adds to the idea mentioned in 1 (acceleration due to gravity...gravity no longer weakens). So, 13 is a pair.****2 is the opening sentence. 'This approach' in 4 refers to $f(R)$ in 2. So, 2413 is the correct sequence.**[🔍 Answer key/Solution](#)

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Q.19 [11594329]

Directions for question (19): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. Because hummingbirds use so much energy, with a heart rate from 450-1300 beats per minute, they must consume several times their body weight in nectar each day.
2. Their heart rate falls to 35-50 beats per minute at night and their body temperature approaches the surrounding temperature.
3. Hummingbirds are the second largest bird family in the New World, with 320 species, and are some of the smallest birds in the world.
4. In addition, hummingbirds conserve energy by falling into a comatose-like state at night.

Solution:

Correct Answer : 3142

 Answer key/Solution

The strongest clue we have in this paragraph is 'in addition' in sentence 4. It adds to the idea of energy which is mentioned in sentence 1. So, 14 is a pair.

2 explains the 'comatose' state mentioned in 4. So, 42 is a pair too.

Sentence 3 has to be the opening sentence because of its scope. So, 3142 is the correct sequence.

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Q.20 [11594329]

Directions for question (20): The passage given below is followed by four summaries. Choose the option that best captures the author's position.

Cost-benefit analysis hinged on an ever-changing calculation of the monetary value of a human life. If a life could be shown to be expensive, regulation could be justified. If not, it would be blocked or scrapped. The EPA, in 2004 – to allow for more lax air pollution regulations – quietly sliced eight percent off their value of human life, and then another three percent in 2008 by deciding to not adjust for inflation.

- 1 ☐ The quantitative measurement of the value of a human life is inherently flawed.
- 2 ☐ Due to the change in EPA regulations, the monetary value of a human life has decreased.
- 3 ☐ Deciding the monetary value of a human life depends upon a multitude of factors, not all of which are justified.
- 4 ☐ Calculating the relative monetary value of a human life doesn't appear to be a fair process.

Solution:

Correct Answer : 4

 Answer key/Solution

The main points of the paragraph are:

Cost-benefit analysis uses a calculation that keeps changing.

The explanation is to show how the calculation is manipulated.

We can eliminate the incorrect options in order to find the answer.

Option 1 – 'Inherently flawed' can't be defined from this paragraph. It is beyond the scope. So, this is not a summary.

Option 2 – It creates a wrong cause-effect relationship.

Option 3 – The concept of 'justified' has not been discussed.

Option 4 – This is the best choice.

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Q.21 [11594329]

Directions for question (21): Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

1. The insulin then helps break down the sugars, which either provide immediate energy or go into fat cells for storage.
2. A few recent studies suggest that consuming fake sugar actually trains your insulin response to store more, not less, fat.
3. If your body interprets something as sweet which is not real sugar, it ends up producing that same insulin response.
4. The main one is that artificial sweeteners are actually better for you than real sugar.
5. Basically when you consume real sugar, your taste-buds send an alert to your pancreas saying that calories are on the way.

Solution:

Correct Answer : 4

 Answer key/Solution

The correct order is 2513. All these sentences talk about how artificial sugars affect the pancreas. The sentences explain the process.

Sentence 4 starts with 'the main one'. It has no connection with any other sentence. So, this is the odd one out.

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Q.22 [11594329]

Directions for question (22): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. I carted it in a wheelbarrow and set it down by the pond.
2. Before leaving, he also showed me how to form them into fist-sized balls.
3. About a week after we started filling the pond, he ordered me to bring out the coil of copper wire from the tool shed.
4. Using a pair of wire-cutters, he snipped the copper into short lengths.

Solution:

Correct Answer : 3142

This is a narrative paragraph. So, the sequence should be easy to decide.

All the sentences use pronouns. So, it is safe to assume that this is part of a larger narrative.

Sentence 3 talks about an order (coil of copper wire). So, it has to be the first sentence. All the other sentences explain the sequence that follows.

1 has the pronoun 'it' which refers to the copper wire.

4 talks about cutting the wire. Sentence 2 has the word 'also'. So 42 is the correct order.

Hence, the correct sequence is 3142.

 Answer key/Solution

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Q.23 [11594329]

Directions for question (23): Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

1. While it supports a sizeable population of wild tigers and other wildlife, it is also an ecologically fragile and climatically vulnerable region that is home to over 4.5 million people.
2. The landscape is constantly being transformed by the erosional forces of the sea and wind along the coast and by the enormous loads of silt and other sediments that are deposited along the myriad estuaries.
3. Short term goals include interventions such as ensuring sustainable livelihoods, access to clean and sustainable energy, and effective human wildlife conflict management.
4. The Sundarbans delta in India has been a priority region for WWF-India since 1973 due to its unique biodiversity.
5. Securing the future of the Sundarbans, its biodiversity, and people requires a long term vision that can integrate climate adaptation and conservation strategies.

Solution:

Correct Answer : 2

The correct order is 4153.

Sentences 5 and 3 talk about strategies to protect the biodiversity of Sundarbans.

Sentence 4 is the opening sentence as it introduces the delta.

So, the odd sentence can be 1 or 2. Sentence 1 talks about biodiversity, the theme of the paragraph.

Sentence 2 talks about the erosion of the landscape. It might be part of the theme, but it doesn't fit into this paragraph. So, 2 is the odd one out.

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 Answer key/Solution

Q.24 [11594329]

Directions for question (24): The passage given below is followed by four summaries. Choose the option that best captures the author's position.

Some analysis of ancient chariots provide that the Egyptians greatly improved the design of this vehicle. However, while they certainly did make improvements to certain parts of the chariot, it is arguable whether the Egyptian chariot was better, or simply designed for a different purpose and terrain than others in the Middle East. For example, the Egyptian chariot had a metal covering for the axes, which reduced friction, and this was certainly an improvement. Also, some wooden parts were strengthened by covering them with metal sleeves. However, the fact that the Egyptian chariots were lighter and faster than those of other major powers in the Middle East may not have been considered an absolute improvement in the chariot's design.

1 ☐ Although the Egyptians improved the overall design of the chariot, it was no match for the Middle Eastern designs.

2 ☐ Research finds that the Egyptians improved the overall design of the ancient chariots; however, questions exist as to the overall efficacy of these.

3 ☐ Some analysts find that the Egyptians improved upon the overall structure of the chariot while others feel that the Egyptians had an unfair advantage in terms of a better terrain and speed.

4 ☐ The wooden parts and metal sleeves made the Egyptians chariots better than the Middle Eastern ones.

Solution:

Correct Answer : 2

The main points of the paragraph are:

The ancient Egyptians improved upon the overall design of chariots.

We can't call these chariots superior to other chariots as some of the design changes (lighter and faster) might not have been considered improvements.

Option 2 covers both the points.

Option 1 – There is no such comparison mentioned in the paragraph.

Option 4 – It is too narrow.

Option 3 – It is misleading. The second part of the option is not mentioned in the paragraph.

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 Answer key/Solution

Q.25 [11594329]

Directions for question (25): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

1. Even if we do manage time away from the grind, it comes with a looming awareness of the things we should be doing.
2. We put off sleeping in, or going for a long walk, or reading by the window.
3. The break that we take, thus, ends up being an experience weighed down by guilt.
4. The problem comes when we spend so long frantically chasing productivity, we refuse to take real breaks.

Solution:

Correct Answer : 4213

Sentence 3 has the linking word 'thus'. So, it is the concluding sentence of the paragraph. 3 defines the effect of not taking real breaks.

4 opens the paragraph by introducing the topic: how we don't take breaks. 2 explains this further.

1 gives a contradiction: how the breaks we take (even if we do) are not real breaks.

So, the correct sequence is 4213.

 Answer key/Solution

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Sec 2

Directions for questions 26 to 29: Answer the questions on the basis of the information given below.

Last year, seven stores—Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech, and Tradelink—each sold exactly 100, 200, 300, 400, or 500 computers, consistent with the following conditions:

1. HP World sold exactly the same number of computers as Rainbow Computers did.
2. Tech Exclusive sold exactly the same number of computers as Tradelink did.
3. Neither Laptopwala.com nor Ansal Infotech sold exactly the same number of computers as any other store.
4. Computer India sold more computers than Tradelink did.
5. Tech Exclusive sold more computers than HP World did.

Q.26 [11594329]

Which one of the following could be an accurate number of computers sold last year by the stores Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech and Tradelink respectively, in the same order?

1 ☐ 100, 200, 300, 400, 200, 500, 300

2 ☐ 100, 200, 400, 300, 200, 500, 400

3 ☐ 500, 200, 300, 400, 500, 100, 300

4 ○ 500, 300, 200, 400, 300, 100, 200



Solution:

Correct Answer : 1

Your Answer : 1

Let computer sold by Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech and Tradelink be a, b, c, d, e, f and g respectively.

By statement 1, $b = e$ and,

By statement 2, $c = g$

By statement 4 and 5, we have, $d > g$ and $c > b$ respectively.

By combining all the relations mentioned above, we get,

$d > c = g > b = e$

Option (1) satisfies all the statements mentioned and the derived relationship.

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Directions for questions 26 to 29: Answer the questions on the basis of the information given below.

Last year, seven stores—Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech, and Tradelink—each sold exactly 100, 200, 300, 400, or 500 computers, consistent with the following conditions:

1. HP World sold exactly the same number of computers as Rainbow Computers did.
2. Tech Exclusive sold exactly the same number of computers as Tradelink did.
3. Neither Laptopwala.com nor Ansal Infotech sold exactly the same number of computers as any other store.
4. Computer India sold more computers than Tradelink did.
5. Tech Exclusive sold more computers than HP World did.

Q.27 [11594329]

How many of the above mentioned stores could have sold exactly 400 computers last year?



[Answer key/Solution](#)
Solution:**Correct Answer : 5****Your Answer : 1**

Let computer sold by Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech and Tradelink be a, b, c, d, e, f and g respectively.

By statement 1, $b = e$ and,

By statement 2, $c = g$

By statement 4 and 5, we have, $d > g$ and $c > b$ respectively.

By combining all the relations mentioned above, we get,

$d > c = g > b = e$

Possible values taken by a, b, c, d, e, f and g could be:

a	b	c	d	e	f	g
400	100	200	300	100	500	200
300	100	200	400	100	500	200
500	100	200	300	100	400	200
200	100	400	500	100	300	400

Except b and e, all other five stores could have sold exactly 400 computers last years.

Note: More cases are possible. Here only those cases are given which are question relevant.

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Directions for questions 26 to 29: Answer the questions on the basis of the information given below.

Last year, seven stores—Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech, and Tradelink—each sold exactly 100, 200, 300, 400, or 500 computers, consistent with the following conditions:

1. HP World sold exactly the same number of computers as Rainbow Computers did.
2. Tech Exclusive sold exactly the same number of computers as Tradelink did.
3. Neither Laptopwala.com nor Ansal Infotech sold exactly the same number of computers as any other store.
4. Computer India sold more computers than Tradelink did.
5. Tech Exclusive sold more computers than HP World did.

Q.28 [11594329]

If a condition that, "Laptopwala.com sold more computers than Ansal Infotech, who sold more computers than Tech Exclusive" is added to the above mentioned conditions, then how many combinations of (store, computers sold) are possible?

[Answer key/Solution](#)
Solution:**Correct Answer : 3**

Let computer sold by Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech and Tradelink be a, b, c, d, e, f and g respectively.

By statement 1, $b = e$ and,

By statement 2, $c = g$

By statement 4 and 5, we have, $d > g$ and $c > b$ respectively.

By combining all the relations mentioned above, we get,

$d > c = g > b = e$

By the condition given in the question,

$a > f > c$

Therefore, the updated relation between the variables will be

$d > c = g > b = e$... (i)

$a > f > c = g > b = e$... (ii)

Possible values the variables can take are:-

a	b	c	d	e	f	g
500	100	200	300	100	400	200
500	100	200	400	100	300	200
400	100	200	500	100	300	200

∴ 3 combinations are possible.

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Directions for questions 26 to 29: Answer the questions on the basis of the information given below.

Last year, seven stores—Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech, and Tradelink—each sold exactly 100, 200, 300, 400, or 500 computers, consistent with the following conditions:

1. HP World sold exactly the same number of computers as Rainbow Computers did.
2. Tech Exclusive sold exactly the same number of computers as Tradelink did.
3. Neither Laptopwala.com nor Ansal Infotech sold exactly the same number of computers as any other store.
4. Computer India sold more computers than Tradelink did.
5. Tech Exclusive sold more computers than HP World did.

Q.29 [11594329]

How many of the following statement/(s) must be true?

- (I) Laptopwala.com sold more computers last year than HP World did.
- (II) Tech Exclusive sold more computers last year than Ansal Infotech did.
- (III) Computer India sold more computers last year than Laptopwala.com did.
- (IV) Tradelink sold more computers last year than Rainbow Computers did.

Enter '0' if none of these statements are true.

Solution:**Correct Answer : 1****Your Answer : 2**[Answer key/Solution](#)

Let computer sold by Laptopwala.com, HP World, Tech Exclusive, Computer India, Rainbow Computers, Ansal Infotech and Tradelink be a, b, c, d, e, f and g respectively.

By statement 1, $b = e$ and,

By statement 2, $c = g$

By statement 4 and 5, we have, $d > g$ and $c > b$ respectively.

By combining all the relations mentioned above, we get,

$d > c = g > b = e$

Only statement IV is definitely true, since $g > e$. (By statement 1, 2 and 4)

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Directions for questions 30 to 33: Answer the questions on the basis of the information given below.

Six students-Praveen, Qureshi, Rohit, Sanjeev, Tarun and Usman-participated in a panel interview of 5 members Akbar, Boman, Chander, Deepak and Ekta -each member could give maximum 10 marks to a student and thus maximum marks that a student could get was 50. Score obtained by the students were 41, 40, 38, 43, 36 and 37 in any order. No member gave less than 6 marks to any student. Neither a member can give 10 marks to more than one student nor a student can get 10 marks from more than one member. The partial information regarding the scores are as below:-

Member \ Student	Akbar	Boman	Chander	Deepak	Ekta
Praveen	7		8	10	
Qureshi		9		7	8
Rohit	6		8		9
Sanjeev	8	8		8	9
Tarun		7	9		7
Usman	10		7	6	
Total	48	46	45	47	49

Q.30 [11594329]

Who got the overall least marks in the interview?

1 ☐ Rohit

2 ☐ Qureshi

3 ☐ Usman

4 ☐ Either (1) or (3)



Solution:

Correct Answer : 4

Your Answer : 3

 **Answer key/Solution**

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	c	8	10	j	
Qureshi	a	9	f	7	8	
Rohit	6	d	8	h	9	
Sanjeev	8	8	g	8	9	
Tarun	b	7	9	i	7	
Usman	10	e	7	6	k	
Total	48	46	45	47	49	

For above table; we have

$$a + b = 48 - (7 + 6 + 8 + 10) = 48 - 31 = 17$$

$$\text{Similarly; } c + d + e = 22, f + g = 13, h + i = 16 \text{ \& } j + k = 16$$

$$\therefore a + b = 17 \therefore a = 8 \text{ or } 9 \text{ and } b = 8 \text{ or } 9$$

[NOTE:- No member can give 10 marks to more than one student.]

$$\therefore f + g = 13 \therefore f = 6 \text{ or } 7 \text{ and } g = 7 \text{ or } 6$$

$$\text{Total score of Sanjeev} = 8 + 8 + 8 + 9 + g = 33 + g$$

$$'g' \text{ must be equal to } 7 \text{ and score of Sanjeev} = 33 + 7 = 40.$$

$$\therefore f = 6$$

$$\text{Now, Score of Qureshi} = 9 + 6 + 7 + 8 + a = 30 + a$$

we can conclude that 'a' must be equal to 8 and score of Qureshi = $30 + 8 = 38$.

$$\therefore b \text{ must be equal to } 9.$$

$$\text{We have; } h + i = 16$$

$$\text{Tarun's Score} = 9 + 7 + 9 + 7 + i = 32 + i$$

As per instruction given in the question; $32 + i$ can be equal to 41 ($i \geq 10$ and $i \leq 6$)

$$\therefore i = 41 - 32 = 9 \text{ and } h = 16 - 9 = 7.$$

$$\text{Score of Rohit} = 6 + d + 8 + 7 + 9 = 30 + d$$

Score of Rohit may be equal to 36 or 37 because 43 can't be score of Rohit as $d \neq 13$

Case I :- Rohit's score = 36 and $d = 6$

$$\text{We have } c + d + e = 22$$

$$c + 6 + e = 22$$

$$c + e = 22 - 6 = 16$$

'e' can't be equal to 10 as Usman and Praveen got 10 marks from Akbar and Deepak respectively.

Possible values of 'c' are 7/8/9.

Total score of Praveen must be 43 because the sum of c and j must be equal to 12 as c is always more than 6.

$$\therefore c + j = 43 - (7 + 8 + 10) = 18.$$

$$\text{We can see that } c = 9 \text{ and } j = 9$$

(because neither 'c' nor 'j' can be equal to 10).

$$\text{Now, } e = 16 - c = 16 - 9 = 7 \text{ and } k = 16 - j = 16 - 9 = 7$$

Final table is as below:-

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	6	8	7	9	36
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	7	7	6	7	37
Total	48	46	45	47	49	

Case II:- Rohit's score = 37 and $d = 7$.

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	7	8	7	9	37
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	6	7	6	7	36
Total	48	46	45	47	49	

Either Rohit or Qureshi got the least marks.

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Directions for questions 30 to 33: Answer the questions on the basis of the information given below.

Six students-Praveen, Qureshi, Rohit, Sanjeev, Tarun and Usman-participated in a panel interview of 5 members Akbar, Boman, Chander, Deepak and Ekta -each member could give maximum 10 marks to a student and thus maximum marks that a student could get was 50. Score obtained by the students were 41, 40, 38, 43, 36 and 37 in any order. No member gave less than 6 marks to any student. Neither a member can give 10 marks to more than one student nor a student can get 10 marks from more than one member. The partial information regarding the scores are as below:-

Member \ Student	Akbar	Boman	Chander	Deepak	Ekta
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Qureshi		9		7	8
Rohit	6		8		9
Sanjeev	8	8		8	9
Tarun		7	9		7
Usman	10		7	6	
Total	48	46	45	47	49

Q.31 [11594329]

Which of the following marks was not given by Boman to any Candidate?

1 ☐ 6

2 ☐ 10

3 ☐ Both (1) and (2)

4 ☐ Either (1) or (2)



Solution:

Correct Answer : 2

Your Answer : 2

[Answer key/Solution](#)

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	c	8	10	j	
Qureshi	a	9	f	7	8	
Rohit	6	d	8	h	9	
Sanjeev	8	8	g	8	9	
Tarun	b	7	9	i	7	
Usman	10	e	7	6	k	
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For above table; we have

$$a + b = 48 - (7 + 6 + 8 + 10) = 48 - 31 = 17$$

$$\text{Similarly; } c + d + e = 22, f + g = 13, h + i = 16 \text{ \& } j + k = 16$$

$$\therefore a + b = 17 \therefore a = 8 \text{ or } 9 \text{ and } b = 8 \text{ or } 9$$

[NOTE:- No member can give 10 marks to more than one student.]

$$\therefore f + g = 13 \therefore f = 6 \text{ or } 7 \text{ and } g = 7 \text{ or } 6$$

$$\text{Total score of Sanjeev} = 8 + 8 + 8 + 9 + g = 33 + g$$

$$'g' \text{ must be equal to } 7 \text{ and score of Sanjeev} = 33 + 7 = 40.$$

$$\therefore f = 6$$

$$\text{Now, Score of Qureshi} = 9 + 6 + 7 + 8 + a = 30 + a$$

we can conclude that 'a' must be equal to 8 and score of Qureshi = $30 + 8 = 38$.

$$\therefore b \text{ must be equal to } 9.$$

$$\text{We have; } h + i = 16$$

$$\text{Tarun's Score} = 9 + 7 + 9 + 7 + i = 32 + i$$

As per instruction given in the question; $32 + i$ can be equal to 41 ($i \geq 10$ and $i \leq 6$)

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'e' can't be equal to 10 as Usman and Praveen got 10 marks from Akbar and Deepak respectively.

Possible values of 'c' are 7/8/9.

Total score of Praveen must be 43 because the sum of c and j must be equal to 12 as c is always more than 6.

$$\therefore c + j = 43 - (7 + 8 + 10) = 18.$$

$$\text{We can see that } c = 9 \text{ and } j = 9$$

(because neither 'c' nor 'j' can be equal to 10).

$$\text{Now, } e = 16 - c = 16 - 9 = 7 \text{ and } k = 16 - j = 16 - 9 = 7$$

Final table is as below:-

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	6	8	7	9	36
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	7	7	6	7	37
Total	48	46	45	47	49	

Case II:- Rohit's score = 37 and $d = 7$.

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	7	8	7	9	37
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	6	7	6	7	36
Total	48	46	45	47	49	

Boman had not give 10 marks to any student.

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Directions for questions 30 to 33: Answer the questions on the basis of the information given below.

Six students-Praveen, Qureshi, Rohit, Sanjeev, Tarun and Usman-participated in a panel interview of 5 members Akbar, Boman, Chander, Deepak and Ekta -each member could give maximum 10 marks to a student and thus maximum marks that a student could get was 50. Score obtained by the students were 41, 40, 38, 43, 36 and 37 in any order. No member gave less than 6 marks to any student. Neither a member can give 10 marks to more than one student nor a student can get 10 marks from more than one member. The partial information regarding the scores are as below:-

Member \ Student	Akbar	Boman	Chander	Deepak	Ekta
Praveen	7		8	10	
Qureshi		9		7	8
Rohit	6		8		9
Sanjeev	8	8		8	9
Tarun		7	9		7
Usman	10		7	6	
Total	48	46	45	47	49

Q.32 [11594329]

How many of the following statement/(s) must be true?

- (I) Boman gave minimum marks to Usman.
- (II) Only two students got 10 marks by the interviewer.
- (III) Praveen got the highest marks.
- (IV) Tarun got the second highest marks.

Enter '0' if none of these statements are true.

✕

Solution:

Correct Answer : 3

Your Answer : 2

[Answer key/Solution](#)

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	c	8	10	j	
Qureshi	a	9	f	7	8	
Rohit	6	d	8	h	9	
Sanjeev	8	8	g	8	9	
Tarun	b	7	9	i	7	
Usman	10	e	7	6	k	
Total	48	46	45	47	49	

For above table; we have

$$a + b = 48 - (7 + 6 + 8 + 10) = 48 - 31 = 17$$

$$\text{Similarly; } c + d + e = 22, f + g = 13, h + i = 16 \text{ \& } j + k = 16$$

$$\therefore a + b = 17 \therefore a = 8 \text{ or } 9 \text{ and } b = 8 \text{ or } 9$$

[NOTE:- No member can give 10 marks to more than one student.]

$$\therefore f + g = 13 \therefore f = 6 \text{ or } 7 \text{ and } g = 7 \text{ or } 6$$

$$\text{Total score of Sanjeev} = 8 + 8 + 8 + 9 + g = 33 + g$$

$$'g' \text{ must be equal to } 7 \text{ and score of Sanjeev} = 33 + 7 = 40.$$

$$\therefore f = 6$$

$$\text{Now, Score of Qureshi} = 9 + 6 + 7 + 8 + a = 30 + a$$

we can conclude that 'a' must be equal to 8 and score of Qureshi = $30 + 8 = 38$.

$$\therefore b \text{ must be equal to } 9.$$

$$\text{We have; } h + i = 16$$

$$\text{Tarun's Score} = 9 + 7 + 9 + 7 + i = 32 + i$$

As per instruction given in the question; $32 + i$ can be equal to 41 ($i \geq 10$ and $i \leq 6$)

$$\therefore i = 41 - 32 = 9 \text{ and } h = 16 - 9 = 7.$$

$$\text{Score of Rohit} = 6 + d + 8 + 7 + 9 = 30 + d$$

Score of Rohit may be equal to 36 or 37 because 43 can't be score of Rohit as $d \neq 13$

Case I :- Rohit's score = 36 and $d = 6$

$$\text{We have } c + d + e = 22$$

$$c + 6 + e = 22$$

$$c + e = 22 - 6 = 16$$

'e' can't be equal to 10 as Usman and Praveen got 10 marks from Akbar and Deepak respectively.

Possible values of 'c' are 7/8/9.

Total score of Praveen must be 43 because the sum of c and j must be equal to 12 as c is always more than 6.

$$\therefore c + j = 43 - (7 + 8 + 10) = 18.$$

$$\text{We can see that } c = 9 \text{ and } j = 9$$

(because neither 'c' nor 'j' can be equal to 10).

$$\text{Now, } e = 16 - c = 16 - 9 = 7 \text{ and } k = 16 - j = 16 - 9 = 7$$

Final table is as below:-

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	6	8	7	9	36
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	7	7	6	7	37
Total	48	46	45	47	49	

Case II:- Rohit's score = 37 and $d = 7$.

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	7	8	7	9	37
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	6	7	6	7	36
Total	48	46	45	47	49	

Statement II, III and IV are true.

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FeedBack

Directions for questions 30 to 33: Answer the questions on the basis of the information given below.

Six students-Praveen, Qureshi, Rohit, Sanjeev, Tarun and Usman-participated in a panel interview of 5 members Akbar, Boman, Chander, Deepak and Ekta -each member could give maximum 10 marks to a student and thus maximum marks that a student could get was 50. Score obtained by the students were 41, 40, 38, 43, 36 and 37 in any order. No member gave less than 6 marks to any student. Neither a member can give 10 marks to more than one student nor a student can get 10 marks from more than one member. The partial information regarding the scores are as below:-

Member \ Student	Akbar	Boman	Chander	Deepak	Ekta
Praveen	7		8	10	
Qureshi		9		7	8
Rohit	6		8		9
Sanjeev	8	8		8	9
Tarun		7	9		7
Usman	10		7	6	
Total	48	46	45	47	49

Q.33 [11594329]

Which of the following can be the sequence representing the name of students getting marks in increasing order?

- 1 ☐ Rohit, Qureshi, Usman, Praveen, Sanjeev, Tarun
- 2 ☐ Rohit, Usman, Qureshi, Sanjeev, Praveen, Tarun
- 3 ☐ Usman, Rohit, Qureshi, Sanjeev, Tarun, Praveen
- 4 ☐ Usman, Rohit, Sanjeev, Qureshi, Tarun, Praveen.



Solution:

Correct Answer : 3

Your Answer : 3

[Answer key/Solution](#)

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	c	8	10	j	
Qureshi	a	9	f	7	8	
Rohit	6	d	8	h	9	
Sanjeev	8	8	g	8	9	
Tarun	b	7	9	i	7	
Usman	10	e	7	6	k	
Total	48	46	45	47	49	

For above table; we have

$$a + b = 48 - (7 + 6 + 8 + 10) = 48 - 31 = 17$$

$$\text{Similarly; } c + d + e = 22, f + g = 13, h + i = 16 \text{ \& } j + k = 16$$

$$\therefore a + b = 17 \therefore a = 8 \text{ or } 9 \text{ and } b = 8 \text{ or } 9$$

[NOTE:- No member can give 10 marks to more than one student.]

$$\therefore f + g = 13 \therefore f = 6 \text{ or } 7 \text{ and } g = 7 \text{ or } 6$$

$$\text{Total score of Sanjeev} = 8 + 8 + 8 + 9 + g = 33 + g$$

$$'g' \text{ must be equal to } 7 \text{ and score of Sanjeev} = 33 + 7 = 40.$$

$$\therefore f = 6$$

$$\text{Now, Score of Qureshi} = 9 + 6 + 7 + 8 + a = 30 + a$$

we can conclude that 'a' must be equal to 8 and score of Qureshi = $30 + 8 = 38$.

$$\therefore b \text{ must be equal to } 9.$$

$$\text{We have; } h + i = 16$$

$$\text{Tarun's Score} = 9 + 7 + 9 + 7 + i = 32 + i$$

As per instruction given in the question; $32 + i$ can be equal to 41 ($i \geq 10$ and $i \leq 6$)

$$\therefore i = 41 - 32 = 9 \text{ and } h = 16 - 9 = 7.$$

$$\text{Score of Rohit} = 6 + d + 8 + 7 + 9 = 30 + d$$

Score of Rohit may be equal to 36 or 37 because 43 can't be score of Rohit as $d \neq 13$

Case I :- Rohit's score = 36 and $d = 6$

$$\text{We have } c + d + e = 22$$

$$c + 6 + e = 22$$

$$c + e = 22 - 6 = 16$$

'e' can't be equal to 10 as Usman and Praveen got 10 marks from Akbar and Deepak respectively.

Possible values of 'c' are 7/8/9.

Total score of Praveen must be 43 because the sum of c and j must be equal to 12 as c is always more than 6.

$$\therefore c + j = 43 - (7 + 8 + 10) = 18.$$

We can see that $c = 9$ and $j = 9$

(because neither 'c' nor 'j' can be equal to 10).

$$\text{Now, } e = 16 - c = 16 - 9 = 7 \text{ and } k = 16 - j = 16 - 9 = 7$$

Final table is as below:-

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	6	8	7	9	36
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	7	7	6	7	37
Total	48	46	45	47	49	

Case II:- Rohit's score = 37 and $d = 7$.

Member Student	Akbar	Boman	Chander	Deepak	Ekta	Total
Praveen	7	9	8	10	9	43
Qureshi	8	9	6	7	8	38
Rohit	6	7	8	7	9	37
Sanjeev	8	8	7	8	9	40
Tarun	9	7	9	9	7	41
Usman	10	6	7	6	7	36
Total	48	46	45	47	49	

Usma, Rohit, Qureshi, Sanjeev, Tarun, Praveen.

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Directions for questions 34 to 37: Answer the questions on the basis of the information given below.

The following table gives the responses of A, B, C and D in a question paper comprising of 6 questions and their scores. The answer to each of the question is either an YES or NO. Each correct response fetches '+5' marks. There are no negative marks for incorrect response or for not attempting.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	?
D	NO	YES	YES	YES	YES	—	15

Note: “—” denote question not attempted.

Q.34 [11594329]

What is the score of C in the exam?

Solution:

Correct Answer : 10

[🔍 Answer key/Solution](#)

In the table given, if we observe the responses and score of B, then we conclude that B must have given one incorrect response so as to fetch the score of 20 marks.

Let B answered Q1 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO

This case is not possible since, A won't be able to score 10, if Q1 is answered wrong by B.

Let B answered Q2 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO
C	NO	YES	—	YES	NO	NO
D	NO	YES	YES	YES	YES	—

This case is also not possible since, D won't be able to score 15, if Q2 is answered wrong by B.

Similarly, if B answered Q3, and Q6 wrong will also be ruled out and then we will be left with just one case, that is, when B answered Q4 wrong.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	10
D	NO	YES	YES	YES	YES	—	15

Answer key therefore is,

Q1 – YES

Q2- NO

Q3 – YES

Q4 – YES

Q5 – YES and,

Q6 - NO

[Note: '—' denote unanswered question and shaded box denote incorrect answer and unshaded box denote correct answer.]

C scored 10 marks in the exam.

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Directions for questions 34 to 37: Answer the questions on the basis of the information given below.

The following table gives the responses of A, B, C and D in a question paper comprising of 6 questions and their scores. The answer to each of the question is either an YES or NO. Each correct response fetches '+5' marks. There are no negative marks for incorrect response or for not attempting.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	?
D	NO	YES	YES	YES	YES	—	15

Note: “—” denote question not attempted.

Q.35 [11594329]

If B's response to Q5 had been an YES, what would have been his score?

Solution:

Correct Answer : 25

 Answer key/Solution

In the table given, if we observe the responses and score of B, then we conclude that B must have given one incorrect response so as to fetch the score of 20 marks.

Let B answered Q1 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO

This case is not possible since, A won't be able to score 10, if Q1 is answered wrong by B.

Let B answered Q2 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO
C	NO	YES	—	YES	NO	NO
D	NO	YES	YES	YES	YES	—

This case is also not possible since, D won't be able to score 15, if Q2 is answered wrong by B.

Similarly, if B answered Q3, and Q6 wrong will also be ruled out and then we will be left with just one case, that is, when B answered Q4 wrong.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	10
D	NO	YES	YES	YES	YES	—	15

Answer key therefore is,

Q1 – YES

Q2- NO

Q3 – YES

Q4 – YES

Q5 – YES and,

Q6 - NO

[Note: '—' denote unanswered question and shaded box denote incorrect answer and unshaded box denote correct answer.]

If B's response would have been an YES to Q6, then B's score would be 25.

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Directions for questions 34 to 37: Answer the questions on the basis of the information given below.

The following table gives the responses of A, B, C and D in a question paper comprising of 6 questions and their scores. The answer to each of the question is either an YES or NO. Each correct response fetches '+5' marks. There are no negative marks for incorrect response or for not attempting.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	?
D	NO	YES	YES	YES	YES	—	15

Note: "—" denote question not attempted.

Q.36 [11594329]

One of the answers in the answer key was wrong. When the scores were corrected with the new answer key there were two with the same scores and the other two also had identical scores. Which question was incorrect in the original answer key?

1 ☐ Q1

2 ☐ Q3

3 ☐ Q4

4 ☐ Q6

Solution:

Correct Answer : 2

 Answer key/Solution

In the table given, if we observe the responses and score of B, then we conclude that B must have given one incorrect response so as to fetch the score of 20 marks.

Let B answered Q1 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO

This case is not possible since, A won't be able to score 10, if Q1 is answered wrong by B.

Let B answered Q2 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO
C	NO	YES	—	YES	NO	NO
D	NO	YES	YES	YES	YES	—

This case is also not possible since, D won't be able to score 15, if Q2 is answered wrong by B.

Similarly, if B answered Q3, and Q6 wrong will also be ruled out and then we will be left with just one case, that is, when B answered Q4 wrong.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	10
D	NO	YES	YES	YES	YES	—	15

Answer key therefore is,

Q1 – YES

Q2- NO

Q3 – YES

Q4 – YES

Q5 – YES and,

Q6 - NO

[Note: '—' denote unanswered question and shaded box denote incorrect answer and unshaded box denote correct answer.]

Let if answer key of Q1 was wrong, that is, Q1 has answer NO, then the revised score of A, B, C and D will be 5, 15, 15, and 20, not the right choice.

Let if answer key of Q3 was wrong, that is, Q3 has answer NO, then the revised score of A, B, C and D will be 15, 15, 10, and 10, so Q3 is the right choice.

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Directions for questions 34 to 37: Answer the questions on the basis of the information given below.

The following table gives the responses of A, B, C and D in a question paper comprising of 6 questions and their scores. The answer to each of the question is either an YES or NO. Each correct response fetches '+5' marks. There are no negative marks for incorrect response or for not attempting.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	?
D	NO	YES	YES	YES	YES	—	15

Note: “—” denote question not attempted.

Q.37 [11594329]

If any answered responses from the table given above is randomly selected, then what is the probability that it is wrong?

1 ☐ 7/20

2 ☐ 4/5

3 ☐ 9/20

4 ☐ 1/2

Solution:

Correct Answer : 3

[🔍 Answer key/Solution](#)

In the table given, if we observe the responses and score of B, then we conclude that B must have given one incorrect response so as to fetch the score of 20 marks.

Let B answered Q1 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO

This case is not possible since, A won't be able to score 10, if Q1 is answered wrong by B.

Let B answered Q2 incorrectly, then,

	Q1	Q2	Q3	Q4	Q5	Q6
A	YES	—	NO	YES	NO	YES
B	YES	NO	YES	NO	—	NO
C	NO	YES	—	YES	NO	NO
D	NO	YES	YES	YES	YES	—

This case is also not possible since, D won't be able to score 15, if Q2 is answered wrong by B.

Similarly, if B answered Q3, and Q6 wrong will also be ruled out and then we will be left with just one case, that is, when B answered Q4 wrong.

	Q1	Q2	Q3	Q4	Q5	Q6	Score
A	YES	—	NO	YES	NO	YES	10
B	YES	NO	YES	NO	—	NO	20
C	NO	YES	—	YES	NO	NO	10
D	NO	YES	YES	YES	YES	—	15

Answer key therefore is,

Q1 – YES

Q2- NO

Q3 – YES

Q4 – YES

Q5 – YES and,

Q6 - NO

[Note: '—' denote unanswered question and shaded box denote incorrect answer and unshaded box denote correct answer.]

Number of answered responses is those which are either answered YES or NO, which is equal to 20. And the number of responses which are incorrect are 9, therefore, if any answered responses from the table given above are randomly selected, then the probability that it is wrong will be 9/20.

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Directions for questions 38 to 40: Answer the questions on the basis of the information given below.

Seven persons – Pankaj, Ranjeet, Tarun, Saket, Teejay, Karan and Arun work in a company whose weekly-off is on a different day among Monday to Sunday. Their daily wages are among Rs. 350, Rs. 450, Rs. 300 and Rs. 400. The additional information is as below:

(I) Ranjeet's daily wage, whose weekly-off is on Tuesday is more than that of Karan which in turn is more than that of Arun and these two have weekly-off on successive days.

(II) Person having weekly-off on Saturday and Thursday have same wages. At most, 2 persons can have same wage. The weekly-off of Karan was on Thursday.

(III) The sum of wages of Pankaj and Tarun is equal to that of persons having weekly-off on Friday and Saturday whose wages are in increasing order.

(IV) Saket's weekly-off is two days before that of Pankaj, whose salary is Rs. 450.

(V) Wage of the person whose weekly-off is on Wednesday has the same wage as Ranjeet.

[Note: Wages of each of Pankaj and Tarun were different from that of persons having weekly-off on Friday and Saturday and each of them have different wages.]

Q.38 [11594329]

How many persons, out of seven, can have minimum wages?

Solution:

Correct Answer : 2

 **Answer key/Solution**

From statement III, we can conclude the following table regarding wages.

	Pankaj	Tarun	Person whose week-off is on Friday	Week-off on Saturday
Case I	Rs. 350/400	Rs. 400/350	300	450
Case II	Rs. 300/450	Rs. 450/300	350	400
Case III	Rs.400	400	350	450

Case (III) cannot be considered because Pankaj and Tarun cannot have equal wage.

From statement (VI), it can be concluded that only case (II) holds, where wage of Pankaj is Rs. 450 and wage of Tarun is Rs. 300.

From statement (II), the wage of person having weekly day off on Saturday and Thursday is same, which will be Rs. 400. Therefore, the table will be as follows:

Person	Week-off	Daily wages
	Mon	
Ranjeet	Tue	x
	Wed	x
Karan	Thu	400
	Fri	350
	Sat	400
	Sun	

Now, from statement (III) and (IV), Saket's weekly day off is two days before that of Pankaj, whose weekly off cannot be on Friday or Saturday. Wage of Ranjeet must be Rs. 450 because his wage must be more than that of Karan.

Therefore, weekly day off of Pankaj must be on Wednesday and of Saket on Monday. Wage of Tarun is Rs. 300 and his weekly day off must be on Sunday. Weekly off of Arun must be on Friday from statement (I).

Therefore, final table will be

Person	Week-off	Daily wages
Saket	Mon	300/350
Ranjeet	Tue	450
Pankaj	Wed	450
Karan	Thu	400
Arun	Fri	350
Teejay	Sat	400
Tarun	Sun	300

Tarun and Saket may have minimum wages.

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Directions for questions 38 to 40: Answer the questions on the basis of the information given below.

Seven persons – Pankaj, Ranjeet, Tarun, Saket, Teejay, Karan and Arun work in a company whose weekly-off is on a different day among Monday to Sunday. Their daily wages are among Rs. 350, Rs. 450, Rs. 300 and Rs. 400. The additional information is as below:

(I) Ranjeet's daily wage, whose weekly-off is on Tuesday is more than that of Karan which in turn is more than that of Arun and these two have weekly-off on successive days.

(II) Person having weekly-off on Saturday and Thursday have same wages. At most, 2 persons can have same wage. The weekly-off of Karan was on Thursday.

(III) The sum of wages of Pankaj and Tarun is equal to that of persons having weekly-off on Friday and Saturday whose wages are in increasing order.

(IV) Saket's weekly-off is two days before that of Pankaj, whose salary is Rs. 450.

(V) Wage of the person whose weekly-off is on Wednesday has the same wage as Ranjeet.

[Note: Wages of each of Pankaj and Tarun were different from that of persons having weekly-off on Friday and Saturday and each of them have different wages.]

Q.39 [11594329]

The weekly-off of Teejay is on

1 ☐ Friday

2 ☐ Saturday

3 ☐ Sunday

4 ☐ Either (1) or (2)

Solution:

Correct Answer : 2

 **Answer key/Solution**

From statement III, we can conclude the following table regarding wages.

	Pankaj	Tarun	Person whose week-off is on Friday	Week-off on Saturday
Case I	Rs. 350/400	Rs. 400/350	300	450
Case II	Rs. 300/450	Rs. 450/300	350	400
Case III	Rs.400	400	350	450

Case (III) cannot be considered because Pankaj and Tarun cannot have equal wage.

From statement (VI), it can be concluded that only case (II) holds, where wage of Pankaj is Rs. 450 and wage of Tarun is Rs. 300.

From statement (II), the wage of person having weekly day off on Saturday and Thursday is same, which will be Rs. 400. Therefore, the table will be as follows:

Person	Week-off	Daily wages
	Mon	
Ranjeet	Tue	x
	Wed	x
Karan	Thu	400
	Fri	350
	Sat	400
	Sun	

Now, from statement (III) and (IV), Saket's weekly day off is two days before that of Pankaj, whose weekly off cannot be on Friday or Saturday. Wage of Ranjeet must be Rs. 450 because his wage must be more than that of Karan.

Therefore, weekly day off of Pankaj must be on Wednesday and of Saket on Monday. Wage of Tarun is Rs. 300 and his weekly day off must be on Sunday. Weekly off of Arun must be on Friday from statement (I).

Therefore, final table will be

Person	Week-off	Daily wages
Saket	Mon	300/350
Ranjeet	Tue	450
Pankaj	Wed	450
Karan	Thu	400
Arun	Fri	350
Teejay	Sat	400
Tarun	Sun	300

The week-off of Teejay is on Saturday.

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Directions for questions 38 to 40: Answer the questions on the basis of the information given below.

Seven persons – Pankaj, Ranjeet, Tarun, Saket, Teejay, Karan and Arun work in a company whose weekly-off is on a different day among Monday to Sunday. Their daily wages are among Rs. 350, Rs. 450, Rs. 300 and Rs. 400. The additional information is as below:

(I) Ranjeet's daily wage, whose weekly-off is on Tuesday is more than that of Karan which in turn is more than that of Arun and these two have weekly-off on successive days.

(II) Person having weekly-off on Saturday and Thursday have same wages. At most, 2 persons can have same wage. The weekly-off of Karan was on Thursday.

(III) The sum of wages of Pankaj and Tarun is equal to that of persons having weekly-off on Friday and Saturday whose wages are in increasing order.

(IV) Saket's weekly-off is two days before that of Pankaj, whose salary is Rs. 450.

(V) Wage of the person whose weekly-off is on Wednesday has the same wage as Ranjeet.

[Note: Wages of each of Pankaj and Tarun were different from that of persons having weekly-off on Friday and Saturday and each of them have different wages.]

Q.40 [11594329]

Which of the following can have maximum wages?

1 ☐ Tarun

2 ☐ Pankaj

3 ☐ Saket

4 ☐ Teejay

Solution:

Correct Answer : 2

 **Answer key/Solution**

From statement III, we can conclude the following table regarding wages.

	Pankaj	Tarun	Person whose week-off is on Friday	Week-off on Saturday
Case I	Rs. 350/400	Rs. 400/350	300	450
Case II	Rs. 300/450	Rs. 450/300	350	400
Case III	Rs.400	400	350	450

Case (III) cannot be considered because Pankaj and Tarun cannot have equal wage.

From statement (VI), it can be concluded that only case (II) holds, where wage of Pankaj is Rs. 450 and wage of Tarun is Rs. 300.

From statement (II), the wage of person having weekly day off on Saturday and Thursday is same, which will be Rs. 400. Therefore, the table will be as follows:

Person	Week-off	Daily wages
	Mon	
Ranjeet	Tue	x
	Wed	x
Karan	Thu	400
	Fri	350
	Sat	400
	Sun	

Now, from statement (III) and (IV), Saket's weekly day off is two days before that of Pankaj, whose weekly off cannot be on Friday or Saturday. Wage of Ranjeet must be Rs. 450 because his wage must be more than that of Karan.

Therefore, weekly day off of Pankaj must be on Wednesday and of Saket on Monday. Wage of Tarun is Rs. 300 and his weekly day off must be on Sunday. Weekly off of Arun must be on Friday from statement (I).

Therefore, final table will be

Person	Week-off	Daily wages
Saket	Mon	300/350
Ranjeet	Tue	450
Pankaj	Wed	450
Karan	Thu	400
Arun	Fri	350
Teejay	Sat	400
Tarun	Sun	300

Pankaj has the maximum wage.

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Directions for questions 41 - 42: Answer the questions on the basis of the information given below.

A committee of 11 members is to be formed by the Prime Minister from a group of 15 key members – K, L, M, N, P, Q, R, S, T, U, V, W, X, Y and Z. Among the members, K, N, V, X, Y and Z are the members of Lok Sabha; L, M, Q, R and S are the members of Rajya Sabha; P and W are the Chief Ministers; T and U are the retired judges of Supreme Court. It is also known that:

- (i) Q can be selected in the committee only if both R and Y are selected.
- (ii) If T is selected, then Q is also selected.
- (iii) V and X cannot be selected together in the committee. The same is true for L and R.
- (iv) The committee must have at least 4 members of Lok Sabha, at least 1 Chief Minister and exactly 1 retired judge.
- (v) S can be selected only if L is selected.

Q.41 [11594329]

If R is one of the members of Rajya Sabha in the committee, then who will be the Chief Minister(s)?

1 ☐ P

2 ☐ W

3 ☐ Either (1) or (2)

4 ☐ Both (1) and (2)

Solution:

Correct Answer : 4

 Answer key/Solution

From statement (iii), L and R cannot be together in the committee. Therefore, there are three possible cases.

Case I: When L is selected.

Q cannot be selected as Q can only be selected when both R and Y are selected. Thus, when L is selected, the committee comprises exactly three members of Rajya Sabha and two Chief Ministers. Also, U must be the judge in the committee, as selection of T ensures selection of Q. Following table gives the possible compositions for the committee.

Members of Lok Sabha	Members of Rajya Sabha	Chief Ministers	Retired Judges
K, N, Y, Z, V/X	L, S, M	P, W	U

The number of ways in which the committee can be formed = $2 \times 1 \times 1 \times 1 = 2$

Case II: When R is selected.

S cannot be selected as S can only be selected when L is selected. Thus, when R is selected, then again the committee comprises exactly three members of Rajya Sabha and two Chief Ministers. Following table gives the possible compositions for the committee.

Members of Lok Sabha	Members of Rajya Sabha	Chief Ministers	Retired Judges
K, N, Y, Z, V/X	R, Q, M	P, W	T/U

The number of ways in which the committee can be formed = $2 \times 1 \times 1 \times 2 = 4$.

Case III: When neither L nor R is selected.

When both L and R are not selected, then there is no possible composition for the committee.

If R is one of the members of Rajya Sabha in the committee, then both of the two (i.e., P and W) can be selected as the Chief Ministers in the committee.

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Directions for questions 41 - 42: Answer the questions on the basis of the information given below.

A committee of 11 members is to be formed by the Prime Minister from a group of 15 key members – K, L, M, N, P, Q, R, S, T, U, V, W, X, Y and Z. Among the members, K, N, V, X, Y and Z are the members of Lok Sabha; L, M, Q, R and S are the members of Rajya Sabha; P and W are the Chief Ministers; T and U are the retired judges of Supreme Court. It is also known that:

(i) Q can be selected in the committee only if both R and Y are selected.

(ii) If T is selected, then Q is also selected.

(iii) V and X cannot be selected together in the committee. The same is true for L and R.

(iv) The committee must have at least 4 members of Lok Sabha, at least 1 Chief Minister and exactly 1 retired judge.

(v) S can be selected only if L is selected.

Q.42 [11594329]

Which of the following statement(s) is/are true?

- I. If L is selected, then T is also selected.
 II. If S is selected, then M is also selected.
 III. P, W and M cannot be selected together in the committee.

1 ☐ Only I

2 ☐ Only II

3 ☐ Only III

4 ☐ Both II and III

Solution:

Correct Answer : 2

 Answer key/Solution

From statement (iii), L and R cannot be together in the committee. Therefore, there are three possible cases.

Case I: When L is selected.

Q cannot be selected as Q can only be selected when both R and Y are selected. Thus, when L is selected, the committee comprises exactly three members of Rajya Sabha and two Chief Ministers. Also, U must be the judge in the committee, as selection of T ensures selection of Q. Following table gives the possible compositions for the committee.

Members of Lok Sabha	Members of Rajya Sabha	Chief Ministers	Retired Judges
K, N, Y, Z, V/X	L, S, M	P, W	U

The number of ways in which the committee can be formed = $2 \times 1 \times 1 \times 1 = 2$

Case II: When R is selected.

S cannot be selected as S can only be selected when L is selected. Thus, when R is selected, then again the committee comprises exactly three members of Rajya Sabha and two Chief Ministers. Following table gives the possible compositions for the committee.

Members of Lok Sabha	Members of Rajya Sabha	Chief Ministers	Retired Judges
K, N, Y, Z, V/X	R, Q, M	P, W	T/U

The number of ways in which the committee can be formed = $2 \times 1 \times 1 \times 2 = 4$.

Case III: When neither L nor R is selected.

When both L and R are not selected, then there is no possible composition for the committee.

Only statement II is true.

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Directions for questions 43 to 45: Answer the questions on the basis of the information given below.

Arbutus College owns exactly four houses that it leases to faculty or students. Of these houses, no two are exactly the same distance from Arbutus campus, and each house is either a student house (occupied entirely by students) or a faculty house (occupied entirely by faculty). The lease length for each house is one, two, or three semesters. The following conditions must hold:

1. No student house has a three-semester lease.
2. At least two houses have longer leases than that of the house closest to campus.
3. Every student house (if there are any) is farther from campus than any faculty house (if there are any).

Q.43 [11594329]

Which one of the following could be a complete and accurate list of the leases of the student houses, ordered from the student house closest to campus to the student house farthest from campus?

- 1 ☐ One semester, one semester
- 2 ☐ Two semesters, three semesters
- 3 ☐ One semester, two semesters, one semester
- 4 ☐ One semester, two semesters, two semesters, one semester

Solution:

Correct Answer : 4

 Answer key/Solution

Let the four houses be at the distance of a, b, c and d such that $a > b > c > d$.

Option (1) is not possible, because of statement (2) i.e., at least two houses have larger leases than that of the house closest to campus and since every student's house is farther from campus than any faculty house, therefore, the distance could be a, b or c, hence, not possible.

Option (2), is not possible because of statement 1.

Option (3) is also not possible again because of statement (2).

Option (4) is possible, since it satisfies all the conditions.

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Directions for questions 43 to 45: Answer the questions on the basis of the information given below.

Arbutus College owns exactly four houses that it leases to faculty or students. Of these houses, no two are exactly the same distance from Arbutus campus, and each house is either a student house (occupied entirely by students) or a faculty house (occupied entirely by faculty). The lease length for each house is one, two, or three semesters. The following conditions must hold:

1. No student house has a three-semester lease.
2. At least two houses have longer leases than that of the house closest to campus.
3. Every student house (if there are any) is farther from campus than any faculty house (if there are any).

Q.44 [11594329]

If the house farthest from campus has a lease longer than that of each of the other houses, then which one of the following could be true?

- 1 ☐ Each faculty house has a two-semester lease.
- 2 ☐ Exactly two houses each have a one-semester lease.
- 3 ☐ Exactly three houses each have a two-semester lease.
- 4 ☐ None of the houses has a one-semester lease.

Solution:**Correct Answer : 2**[Answer key/Solution](#)

Let the four houses be at the distance of a, b, c and d such that $a > b > c > d$.

The house farthest from campus has the distance 'a' and have the lease longer than that of each of the other houses, that means lease of the house farthest from campus is three semesters and therefore by statement (3), we get the case where in none of the houses are of students.

∴ Only option (2), is the valid scenario and could be true.

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Directions for questions 43 to 45: Answer the questions on the basis of the information given below.

Arbutus College owns exactly four houses that it leases to faculty or students. Of these houses, no two are exactly the same distance from Arbutus campus, and each house is either a student house (occupied entirely by students) or a faculty house (occupied entirely by faculty). The lease length for each house is one, two, or three semesters. The following conditions must hold:

1. No student house has a three-semester lease.
2. At least two houses have longer leases than that of the house closest to campus.
3. Every student house (if there are any) is farther from campus than any faculty house (if there are any).

Q.45 [11594329]

What is the maximum number of houses that could all be student houses with two-semester leases?

1 ☐ Zero

2 ☐ One

3 ☐ Two

4 ☐ Three

Solution:**Correct Answer : 4**[Answer key/Solution](#)

Let the four houses be at the distance of a, b, c and d such that $a > b > c > d$.

Since, student's house can have a lease of either one-semester or two semester.

Therefore, maximum 3 students houses can be there with two-semester leases because of statement 2 and leases of those houses can be in the combination of (2, 2, 2 and 1).

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Sec 3

Q.46 [11594329]

Two cars, an Alto and a Swift, start at the same time in opposite directions from two distinct points P and Q. Starting from P, the Alto reaches Q in 6 hours 20 minutes and the Swift starting from Q, reaches P in 4 hours 45 minutes. What is the speed of the Swift, if the speed of the Alto is 60 km/ hr?

1 ☐ 110 km/hr2 ☐ 100 km/hr3 ☐ 90 km/hr4 ☐ 80 km/hr**Solution:****Correct Answer : 4****Your Answer : 4**

When distance is same, speed is inversely proportional to time taken to travel.

$$6 \text{ hours } 20 \text{ minutes} = \frac{19}{3} \text{ hours.}$$

$$4 \text{ hours } 45 \text{ minutes} = \frac{19}{4} \text{ hours.}$$

$$\text{Speed of the Swift} = \left(\frac{\frac{19}{3}}{\frac{19}{4}} \right) \times 60 = 80 \text{ km/hr.}$$

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[🔍 Answer key/Solution](#)
Q.47 [11594329]

A vendor purchased 20 items at Rs. 400 per item. He sold 'n' items at n% profit and rest at (100 – n)% loss. Find the number of items sold at profit if he had made overall loss of Rs. 800 in this transaction.

1 ☐ 82 ☐ 153 ☐ 124 ☐ 10

Solution:**Correct Answer : 2****Your Answer : 1**[Answer key/Solution](#)

According to the question,

$$(20 - n) \times \frac{(100 - n)}{100} \times 400 - n \times \frac{n}{100} \times 400 = 800$$

$$\Rightarrow (2000 - 120n + n^2)4 - 4n^2 = 800$$

$$\Rightarrow 8000 - 800 = 480n$$

$$\Rightarrow n = \frac{7200}{480} = 15$$

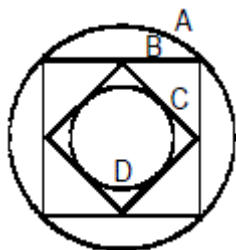
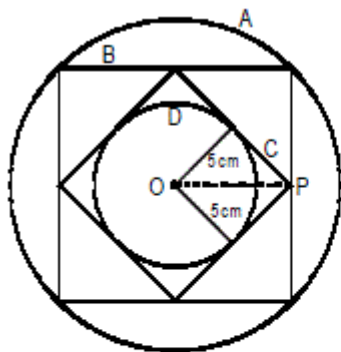
Hence, 15 items were sold at 15% profit.

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Q.48 [11594329]

In this figure, A and D are circles and B and C are squares. Given that the radius of circle D is 5 cm, what will be the sum (in cm) of diameter of circle A and side of square C?

**Solution:****Correct Answer : 30**[Answer key/Solution](#)

Radius of circle D = 5 cm. So, length of side of square C = 10 cm.

$$OP = \sqrt{2}(\text{radius of D}) = 5\sqrt{2}$$

$$\text{So, length of side of square B} = 5\sqrt{2} \times 2 = 10\sqrt{2} \text{ cm}$$

$$\text{Diameter of circle A} = \text{Diagonal of square B} = 10\sqrt{2} \times \sqrt{2} = 20 \text{ cm.}$$

So, the required sum = diameter of circle A + side of square C = 20 + 10 = 30 cm.

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Q.49 [11594329]

Find the 100th term of the sequence $\frac{1}{2}, \frac{5}{3}, \frac{11}{4}, \frac{19}{5}, \dots$

1 ☐ 992 ☐ $\frac{10100}{101}$ 3 ☐ $\frac{9899}{100}$ 4 ☐ $\frac{10099}{101}$ **Solution:****Correct Answer : 4**

$$\frac{1}{2}, 1\frac{2}{3}, 2\frac{3}{4}, 3\frac{4}{5}, \dots, 98\frac{99}{100}, 99\frac{100}{101}$$

$$100\text{th term} = 99\frac{100}{101} = \frac{10099}{101}.$$

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 Answer key/Solution**Q.50 [11594329]**

Four typists A, B, C and D can type atleast 96 pages in 12 days working in turns every four days. The number of pages typed in a day by these four typists working together must be more than

1 ☐ 312 ☐ 323 ☐ 334 ☐ 34

Solution:**Correct Answer : 1****Your Answer : 2**[Answer key/Solution](#)

Let the number of minimum pages types by A, B, C and D in a day be a, b, c and d respectively.

According to the question,

$$a + b + c + d + a + b + c + d + a + b + c + d = 96$$

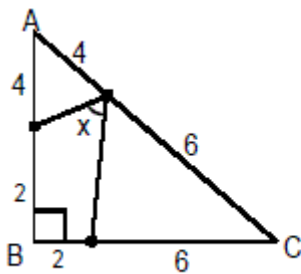
We have to find the minimum value of $a + b + c + d$.

The value of $a + b + c + d$ will be minimum only when $a = b = c = d$ i.e., 8

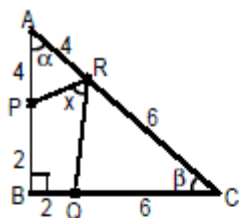
∴ The number of pages typed by four typists together in a day will be $4 \times 8 = 32$ i.e., more than 31.

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Q.51 [11594329]

In $\triangle ABC$ right angled at B, what is the value of x (in degrees)?

Solution:**Correct Answer : 45**[Answer key/Solution](#)In $\triangle ABC$,let $\angle BAC = \alpha^\circ$ and $\angle ACB = \beta^\circ$ $\triangle ARP$ is an isosceles triangle,

$$\therefore \angle ARP = 90^\circ - \frac{\alpha}{2}$$

Similarly, $\triangle RQC$ is an isosceles triangle,

$$\therefore \angle QRC = 90^\circ - \frac{\beta}{2}$$

Now, $\angle ARP + x^\circ + \angle QRC = 180^\circ$

$$90^\circ - \frac{\alpha}{2} + x + 90^\circ - \frac{\beta}{2} = 180^\circ$$

$$x = \frac{\alpha + \beta}{2} \quad \dots (i)$$

In $\triangle ABC$,

$$\alpha + 90^\circ + \beta = 180^\circ$$

$$\therefore \alpha + \beta = 90^\circ \quad \dots (ii)$$

From (i) and (ii),

$$x = \frac{90}{2} = 45^\circ.$$

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Q.52 [11594329]If $\log_a b - \log_b a^{12} = 4$, which of the following must be true?

1 ☐ $a^2 b = 1$

2 ☐ $a = b^6$

3 ☐ $a = b^2$

4 ☐ $ab^2 = 1$

Solution:**Correct Answer : 1**[Answer key/Solution](#)Let $\log_a b = x$

$$\Rightarrow x - \frac{12}{x} = 4 \Rightarrow x^2 - 4x - 12 = 0 \Rightarrow (x - 6)(x + 2) = 0$$

$$\therefore x = \log_a b = 6 \text{ or } \log_a b = -2$$

$$\Rightarrow a^6 = b \text{ or } (a)^{-2} = b \Rightarrow a^2 b = 1.$$

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Q.53 [11594329]

The profit earned on an article when selling price is Rs. 6875 is 14.28% less than the loss occurred when selling price is Rs. 4275. Find the approximate cost price (in Rs.) of the article.

Solution:**Correct Answer : 5675**[🔍 Answer key/Solution](#)

Let the cost price of an article be Rs. 'x'.
According to the question,

$$\Rightarrow 6875 - x = (x - 4275) - \frac{14.28}{100}(x - 4275)$$

$$\Rightarrow 6875 - x = 0.8572(x - 4275)$$

$$\Rightarrow 1.8572x = 10539.53$$

$$\Rightarrow x = \text{Rs. } 5675.$$

[Bookmark](#)[FeedBack](#)**Q.54 [11594329]**

If $f(x) = \frac{1}{x-1} - \frac{1}{x}$ and $g(x) = \frac{x}{3} \left(\frac{1}{x-1} - \frac{1}{x+2} \right)$ for all $x > 0$; the value of $f(g(x))$, at the value of x when $f(x) = g(x)$, will be

1 ☐ 22 ☐ 43 ☐ -24 ☐ -4

Solution:**Correct Answer : 4**[Answer key/Solution](#)

$$f(x) = \frac{1}{x-1} - \frac{1}{x} = \frac{x - (x-1)}{x(x-1)} = \frac{1}{x(x-1)}$$

$$g(x) = \frac{x}{3} \left(\frac{1}{x-1} - \frac{1}{x+2} \right) = \frac{x}{3} \left(\frac{x+2-x+1}{(x-1)(x+2)} \right) = \frac{x}{(x-1)(x+2)}$$

$$\therefore f(x) = g(x) \Rightarrow \frac{1}{x(x-1)} = \frac{x}{(x-1)(x+2)}$$

$$\Rightarrow x^2 = x+2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0 \Rightarrow x = 2 \text{ or } x = -1$$

But, since $x > 0$; $x = 2$

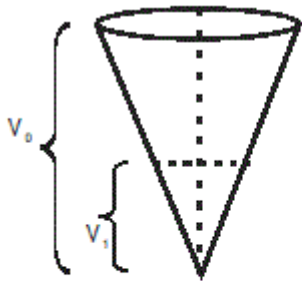
$$\text{At } x = 2; f(g(x)) = f(g(2)) = f\left(\frac{2}{1 \times 4}\right) = f\left(\frac{1}{2}\right) = \frac{1}{\frac{1}{2} \left(\frac{1}{2} - 1 \right)} = \frac{2 \times 2}{1 \times -1} = -4.$$

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Q.55 [11594329]

A child consumed an ice-cream of inverted right-circular conical shape from the top and left only 12.5% of the cone for her mother. If the height of the ice-cream cone was 8 cm, what was the height (in cm) of the remaining ice-cream cone?

Solution:**Correct Answer : 4**[Answer key/Solution](#)

Let total volume of the bigger cone be V_0 .

Total volume of the remaining cone is V_1 .

And let their heights be h_0 and h_1 respectively.

$$\therefore \frac{V_1}{V_0} = \left(\frac{h_1}{h_0} \right)^3 = \frac{1}{8} \Rightarrow h_1 = \frac{h_0}{2} = \frac{8}{2} = 4 \text{ cm}.$$

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Q.56 [11594329]

What is the remainder if $x^{99} + x^{51} + x^{18} + x^3 + 5$ is divided by $x^3 - 1$?

1 ○ 9

2 ○ 1

3 ○ $x^2 + 1$

4 ○ 0

**Solution:****Correct Answer : 1****Your Answer : 1**
[🔍 Answer key/Solution](#)

$$\text{Remainder} \left(\frac{x^{99} + x^{51} + x^{18} + x^3 + 5}{x^3 - 1} \right) = \text{Rem} \left(\frac{(x^{99} - 1) + (x^{51} - 1) + (x^{18} - 1) + (x^3 - 1) + 5 + 4}{x^3 - 1} \right)$$

∴ $(a^n - 1)$ is divisible by $(a - 1)$ if n is odd.

∴ Remainder = 9.

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Q.57 [11594329]

Find the number of different signals that can be generated by arranging at least 2 flags in order (one below the other) on a vertical staff, if five different flag are available.

Solution:**Correct Answer : 320**
[🔍 Answer key/Solution](#)

Since a signal may consist of either 2 flags, 3 flags, 4 flags or 5 flags,
therefore, total number of signals = Number of 2 flags signals + Number of 3 flags signals + Number of 4 flags signals + Number of 5 flags signals
 $= 5 \cdot 4 + 5 \cdot 4 \cdot 3 + 5 \cdot 4 \cdot 3 \cdot 2 + 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$
 $= 20 + 60 + 120 + 120 = 320.$

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Q.58 [11594329]

If $x = \frac{3 + \sqrt{5}}{2}$ and $y = x^3$, then y satisfies the quadratic equation

1 ○ $y^2 - 18y + 1 = 0$ 2 ○ $y^2 + 18y + 1 = 0$ 3 ○ $y^2 - 18y - 1 = 0$

$$4 \bigcirc y^2 + 18y - 1 = 0$$

Solution:

Correct Answer : 1

 Answer key/Solution

$$x = \frac{3 + \sqrt{5}}{2} \Rightarrow 2x - 3 = \sqrt{5}$$

$$\Rightarrow 4x^2 - 12x + 9 = 5 \text{ (squaring both sides)}$$

$$\Rightarrow 4x^2 - 12x + 4 = 0$$

$$\Rightarrow x^2 - 3x + 1 = 0$$

$$\text{Now } x = y^{\frac{1}{3}}$$

$$\Rightarrow y^{\frac{2}{3}} - 3y^{\frac{1}{3}} + 1 = 0$$

$$\Rightarrow y^{\frac{2}{3}} - 3y^{\frac{1}{3}} = -1 \text{ (cubing both sides)}$$

$$\Rightarrow y^2 - 27y - 3y^{\frac{2}{3}} \cdot 3y^{\frac{1}{3}} \left(y^{\frac{2}{3}} - 3y^{\frac{1}{3}} \right) = -1$$

$$\Rightarrow y^2 - 27y - 9y(-1) + 1 = 0$$

$$\Rightarrow y^2 - 18y + 1 = 0.$$

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Q.59 [11594329]

Tank 'A' is filled in four hours when 2 inlet and 3 outlet pipes are opened together and tank 'B' of double capacity as that of A, is filled in 3 hours when 5 inlet and 5 outlet pipes are opened together. If only inlet pipes connected with tank 'A' are operational, then the tank will be filled in [Note: each inlet pipe of both the tanks has same efficiency whereas each outlet pipe of both the tanks have same efficiencies.]

1 ☐ $3\frac{1}{3}$ hours

2 ☐ $2\frac{2}{3}$ hours

3 ☐ 3 hours

4 ☐ 2 hours

Solution:**Correct Answer : 1** Answer key/Solution

Let the flow rates of inlet pipe be 'a' and of outlet pipe be 'b'.

Capacity of Tank A = $4(2a - 3b)$

Capacity of Tank B = $3(5a - 5b)$

Now, $3(5a - 5b) = 2 \times 4(2a - 3b)$

$\Rightarrow 15a - 15b = 16a - 24b \Rightarrow 9b = a$

Also, time required for Tank A = $\frac{4(2a - 3b)}{2a} = \frac{15b \times 4}{18b} = \frac{60}{18} = 3\frac{1}{3}$ hr.

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Q.60 [11594329]

a, b and c are three distinct positive integers less than 25. If $|a - c| + |b - a| = |c - b|$, then the minimum possible value of a is

Solution:**Correct Answer : 2** Answer key/Solution

$|a - c|$ is the distance between two integers a and c on the number line. Similarly, $|b - a|$ and $|c - b|$ are distance between a and b, and b and c respectively.

We can observe that 'a' must be in between b and c on the number line.

$\therefore$ Minimum possible value of a = 2.

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Q.61 [11594329]

Sum of first 10 terms of an A.P. is 100 and that of next 10 terms is 120. Find the first term of the A.P.

1 ☐ 92 ☐ 89/103 ☐ 91/104 ☐ 19/2

Solution:**Correct Answer : 3**[Answer key/Solution](#)

Let the first term be 'a' and common difference be 'd'.

$$\text{Sum of first 10 terms} = \frac{10}{2}[2a + (10-1)d] = 5[2a + 9d] = 100$$

$$\Rightarrow 10a + 45d = 100 \quad \dots (i)$$

$$\text{Sum of next 10 terms} = (a + 10d) + (a + 11d) + \dots + (a + 19d)$$

$$\Rightarrow 10a + 145d = 120 \quad \dots (ii)$$

From equation (i) and (ii),

$$145d - 45d = 120 - 100 = 20$$

$$\Rightarrow 100d = 20 \Rightarrow d = 1/5$$

$$\text{If } d = 1/5, \text{ then } a = \frac{100 - 45d}{10} = \frac{100 - \left(45 \times \frac{1}{5}\right)}{10} = \frac{91}{10}.$$

Hence, the first term is $\frac{91}{10}$.

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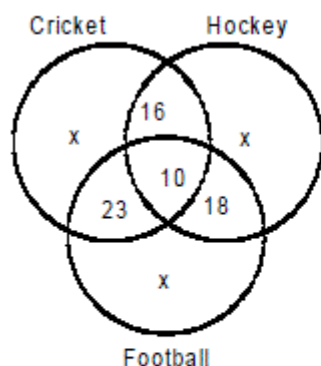
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Q.62 [11594329]

In a survey of 800 students, it was found that 12% of students don't play Cricket, Hockey or Football. 10% play all the three. There are 16% who play only Cricket and Hockey, 18% who play only Hockey and Football and 23% who play only Cricket and Football. Number of students that play only Cricket, only Hockey and only Football are equal. How many students play either Football or Cricket or both?

Solution:**Correct Answer : 648**[Answer key/Solution](#)

Let the percentage of students that play only Cricket, only Hockey, only Football be x each.



$$\text{Then, } x + x + x + 10 + 16 + 18 + 23 + 12 = 100$$

$$\text{Therefore, } 3x = 21 \Rightarrow x = 7$$

$$\text{So, the percentage of students that play either Football or Cricket or both} = 23 + 18 + 10 + 16 + 7 + 7 = 81$$

$$\text{Hence, the number of students that play either Football or Cricket or both} = 800 \times \frac{81}{100} = 648.$$

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Q.63 [11594329]

In a club, red juice and yellow juice were consumed in the ratio of 9 : 4 on the first day. On the second day the juices were consumed in the ratio 2 : 1 respectively. If the ratio of yellow juice consumed on the first day to the yellow juice consumed on the second day was 1 : 2, then what is the minimum number of drinks consumed on both the days? Given that at least 16 drinks of yellow juice were consumed on at least one of the days.

1 ☐ 1482 ☐ 743 ☐ 1814 ☐ 144**Solution:****Correct Answer : 2****Your Answer : 2**
[🔍 Answer key/Solution](#)

Let the red juice and yellow juice consumed on first day be $9x$ and $4x$.

On second day, juices consumed was $16x$ and $8x$ respectively.

To find the minimum drinks consumed, $8x = 16 \Rightarrow x = 2$

Total drinks consumed = $9x + 4x + 16x + 8x = 37x \Rightarrow 37 \times 2 = 74$.

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Q.64 [11594329]

$10^{\log(2-x)} = \log_5(10 - 5^x)$, solve for x .

Solution:**Correct Answer : 1**
[🔍 Answer key/Solution](#)

Applying log to the both sides,

$$\Rightarrow \log(10^{\log(2-x)}) = \log(\log_5(10 - 5^x))$$

$$\Rightarrow \log(2-x) = \log(\log_5(10 - 5^x))$$

$$\Rightarrow (2-x) = \log_5(10 - 5^x)$$

$$\Rightarrow 5^{(2-x)} = 10 - 5^x$$

$$\Rightarrow \frac{5^2}{5^x} = 10 - 5^x$$

$$\Rightarrow \text{put } 5^x = t$$

$$\Rightarrow 25 = (10 - t)t$$

$$\Rightarrow 25 = 10t - t^2$$

$$\Rightarrow t^2 - 10t + 25 = 0$$

$$\Rightarrow (t-5)^2 = 0$$

$$\Rightarrow t = 5$$

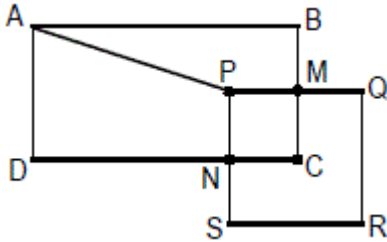
$$\text{i.e., } 5^x = 5, x = 1.$$

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Q.65 [11594329]

ABCD is a rectangle and PQRS is a square as shown in the figure. M is a mid-point of both BC and PQ where $BC = PQ$. The ratio of lengths of CN and ND is 3 : 8. If the area of square PQRS is 144 sq. cm, then find the length of AP (in cm).



1 ☐ $5\sqrt{23}$

2 ☐ $\sqrt{73}$

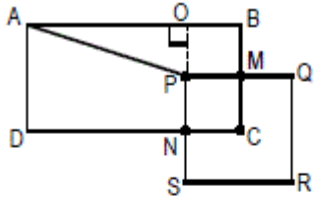
3 ☐ $7\sqrt{23}$

4 ☐ $2\sqrt{73}$

Solution:

Correct Answer : 4

[Answer key/Solution](#)



Let the side of square PQRS be x cm,
 $\therefore$ Area = $x^2 = 144$ sq. cm
 $\therefore x = 12$ cm

$$BM = MC = PM = MQ = \frac{x}{2} = \frac{12}{2} = 6 \text{ cm.}$$

$$OP = OB = NC = PM = 6 \text{ cm}$$

$$\frac{CN}{ND} = \frac{3}{8} \text{ (given)}$$

$$\therefore ND = \frac{8}{3} \times 6 = 16 \text{ cm}$$

$$\therefore AO = 16 \text{ cm}$$

$$\text{In } \triangle AOP, AP^2 = AO^2 + OP^2$$

$$AP = \sqrt{(16)^2 + (6)^2} = 2\sqrt{73} \text{ cm.}$$

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Q.66 [11594329]

Three jugs I, II and III each contains 200 ml of liquids X, Y and Z respectively. 50 ml of X from jug- I is added to jug-II. Then 50 ml of this mixture is added to jug-III. Then 50 ml of this mixture from jug- III is added to jug-I. Finally 100 ml of this mixture from jug I was added to 100 ml of W to prepare a drink. What is the ratio of W : X : Z in the final mixture?

1 ☐ 15 : 19 : 82 ☐ 25 : 19 : 13 ☐ 19 : 5 : 14 ☐ 25 : 19 : 5**Solution:****Correct Answer : 4** Answer key/Solution

All quantities (in ml)	I	II	III	IV
Initial	200(X)	200(Y)	200(Z)	100(W)
First Mixing	150(X)	200(Y) + 50(X)	200(Z)	100(W)
Second Mixing	150(X)	160(Y) + 40(X)	200(Z) + 40(Y) + 10(X)	100(W)
Third Mixing	152(X) + 40(Z) + 8(Y)	160(Y) + 40(X)	160(Z) + 32(Y) + 8(X)	100(W)
Final Mixing	76(X) + 20(Z) + 4(Y)	160(Y) + 40(X)	160(Z) + 32(Y) + 8(X)	100(W) + 76(X) + 20(Z) + 4(Y)

In the final mixture there is 100 ml of W, 76 ml of X, 20 ml of Z and 4 ml of Y.

Hence, required ratio = 100 : 76 : 20 = 25 : 19 : 5.

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Q.67 [11594329]

Set A = {1, 2, 3}. All subsets of A are found except the null set and then the product of the reciprocals of all the elements of all the subsets of set A is calculated. The sum of all these products is

1 ☐ 32 ☐ 43 ☐ 14 ☐ None of these

Solution:**Correct Answer : 1**[Answer key/Solution](#)Set $A = \{1, 2, 3\}$ Subsets of A except null set = $\{\{1\}, \{2\}, \{3\}, \{2, 3\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$ Product of the reciprocals of all the elements of all the subsets of set A is calculated i.e.,

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{2.3}, \frac{1}{1.2}, \frac{1}{1.3}, \frac{1}{1.2.3}$$

$$\text{Sum of all these products} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{2.3} + \frac{1}{1.2} + \frac{1}{1.3} + \frac{1}{1.2.3} = \frac{4! - 3!}{3!} = \frac{3![4 - 1]}{3!} = 3.$$

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Q.68 [11594329]

A cistern has an inlet pipe A and an outlet pipe B. If pipe B is closed, then pipe A fills the empty cistern in 18 minutes. If pipe B is open, then pipe A fills the empty cistern in 24 minutes. If only pipe B is open, then in how many minutes the full cistern becomes three-fourth empty?

1 ☐ 182 ☐ 363 ☐ 544 ☐ 72**Solution:****Correct Answer : 3**[Answer key/Solution](#)

Let the total volume of the cistern be LCM of 18, 24 i.e., 72 units.

Three-fourths empty cistern means $72 \times \frac{3}{4} = 54$ units.In one minute, pipe A fills = $\frac{72}{18} = 4$ units and pipes A and B fill = $\frac{72}{24} = 3$ unitsTherefore, pipe B empties in one minute = $4 - 3 = 1$ unitHence, the time taken by pipe B to make the full cistern three-fourths empty = $\frac{54}{1} = 54$ minutes.

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Q.69 [11594329]

How many integers values of x satisfy the inequality $\frac{x+1}{2x-3} < \frac{x-1}{2x+5}$ where $x \in [-10, 10]$?

Solution:**Correct Answer : 10**[Answer key/Solution](#)

$$\frac{x+1}{2x-3} - \frac{x-1}{2x+5} < 0 \Rightarrow \frac{2x^2+7x+5-2x^2+5x-3}{(2x-3)(2x+5)} < 0 \Rightarrow \frac{12x+2}{(2x-3)(2x+5)} < 0$$

Multiply numerator and denominator by $12x + 2$;

$$\Rightarrow \frac{(12x+2)^2}{(2x-3)(2x+5)(12x+2)} < 0$$

 $\therefore (12x+2)^2$ is always +ve; therefore, $(2x-3)(2x+5)(12x+2)$ must be -ve.

$$\therefore (2x-3)(2x+5)(12x+2) < 0$$

$$\text{So, we have } x \in \left(-\infty, -\frac{5}{2}\right) \cup \left(\frac{-1}{6}, \frac{3}{2}\right).$$

 $\therefore$ values of $x = -10, -9, -8, -7, -6, -5, -4, -3, 0, 1$.

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Q.70 [11594329]

In a mixture of milk and water, there is 24% water. After replacing 6 litres of the mixture with 6 litres of pure milk, the percentage of milk in the mixture became 80%. Find the volume of milk in the original mixture.

1 ☐ 22.04 litres2 ☐ 22.80 litres3 ☐ 21.28 litres4 ☐ 27.36 litres**Solution:****Correct Answer : 4**[Answer key/Solution](#)Let the total mixture be k litres.

According to the question,

$$(100 - 80)\% = 24\% \left(1 - \frac{6}{k}\right) \Rightarrow \frac{20}{24} = 1 - \frac{6}{k} \Rightarrow 1 - \frac{6}{k} = \frac{5}{6}$$

$$\Rightarrow \frac{6}{k} = \frac{1}{6} \therefore k = 36 \text{ litres.}$$

$$\text{Hence, quantity of milk in the original mixture} = 36 \times \frac{(100 - 24)}{100} = 27.36 \text{ litres.}$$

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